In-Context Gibbs: Learning from Weak Labels with Frozen Tabular Foundation Models

Supplementary Material

[TODO: author names]

This document contains all appendices of the paper. References of the form "Section 6.5", "Theorem 3" or "Table 2" point to the main paper; tables, figures, equations and statements numbered with the prefix S, and appendices A, B, . . . , are in this document.

APPENDIX A
PROOFS

Notation.: The main symbols are: X, y, W the features, latent labels and weak labels; ℓ the weak-label likelihood; p the exact posterior predictive (PPD) and $\tilde{p}$ the TFM's approximation of it; π the target posterior over y; K the number of blocks, T, T_0 and S the number of sweeps, the burn-in and the number of samples; F the free points, $n_F = |F|$, $\rho = n_F/n$; r the label accuracy and R_s its value after sweep s; Φ the response map and $g(r) = \Phi(r) - r$ its gain; L the tuned learner of the learner start. Some letters are reused locally, always with a definition: q is the probability that a wrong class enters a candidate set; δ is the tolerance of Assumption 1 and ϑ Dobrushin's coefficient; Δ_1, Δ_2 are the per-dataset differences of Study 3; p is also the failure probability $p(\nu, \lambda)$ of Theorem 3 and a p-value in Section 6; H counts sweeps in Theorem 3, while (H_δ) names the assumption and H1–H5, H2′–H4′ and H1″–H3″ the hypotheses of the three studies; τ is the integrated coefficient of Theorem 2 and a first-passage time in Theorem 3; λ is a margin in Theorem 3, the labelled fraction N/n in Proposition S3 and λ_k a slope in Proposition S5; a is $1 - \lceil n/K \rceil / n_F$ in Theorem 3 and a drift coefficient in Proposition S3; S is also a scoring rule in the proof of Proposition S1, S_i a candidate set, and $S = nR_s$ in the proof of Proposition S4.

Assumptions.: (A0) $W \sim p(W \mid X, y)$, independent of (x_*, y_*) given (X, y); nuisance parameters of the observation model, if any, are a priori independent of (ϕ, X, y). (A1) $\tilde{p}(\cdot \mid x, D) = p(\cdot \mid x, D)$ for every context the algorithm queries. (A2) $p(k \mid x, D) > 0$ for all k and every context of positive probability. (A3) ℓ is known and $\sum_y p(y \mid X)\ell(y) > 0$. For instance-level observations $\ell(y) = \prod_i \ell_i(y_i)$; the set of free points is $F = \{i : |\{k : \ell_i(k) > 0\}| \geq 2\}$.

A.1 Lemma 1

Proof of Lemma 1. Given ϕ the pairs (x_j, y_j) are i.i.d., hence exchangeable, so the joint law of $(D_{-i}, (x_i, y_i))$ equals that of $(D_{n-1}, (x_*, y_*))$, a context of $n-1$ points and a query. The information (X, y_{-i}) coincides with (x_i, D_{-i}). Therefore

$$p(y_i \mid y_{-i}, X) = \frac{p((x_i, y_i), D_{-i})}{\sum_k p((x_i, k), D_{-i})} = p(y_i \mid x_i, D_{-i}). \qquad \square$$

The PPD conditions on the query features x_* as well, which may carry information about ϕ; this is the quantity the PFN training loss targets, since x_* is drawn from the prior together with the context [1].

Corollary S1 (Positivity). *Under (A2), $p(y \mid X) > 0$ for every $y \in [C]^n$.*

Proof. Some y^0 has $p(y^0 \mid X) > 0$. Change one coordinate at a time, $y^0 \to y^1 \to \cdots \to y^m = y$, step k changing point s_k. If $p(y^{k-1} \mid X) > 0$, then $p(y^{k-1}_{-s_k} \mid X) > 0$, the context $D^{k-1}_{-s_k}$ has positive probability, and by Lemma 1 and (A2) $p(y^k \mid X) = p(y^{k-1}_{-s_k} \mid X)\, p(y^k_{s_k} \mid x_{s_k}, D^{k-1}_{-s_k}) > 0$. Induct on k. $\square$

A.2 Theorem 1

Proof of Theorem 1(a). The update of point i is $P_i(y, y') = \pi_i(y'_i \mid y_{-i}) \mathbb{1}[y'_{-i} = y_{-i}]$ with $\pi_i(k \mid y_{-i}) \propto \tilde{p}(k \mid x_i, D_{-i})\ell_i(k)$. By Lemma 1 and (A1), $\pi_i(k \mid y_{-i}) \propto p(k \mid y_{-i}, X)\ell_i(k) \propto \pi(k, y_{-i})$, the full conditional of π; hence $\pi(y)P_i(y, y') = \pi(y_{-i})\pi_i(y_i \mid y_{-i})\pi_i(y'_i \mid y_{-i})$ is symmetric in (y, y'). By Corollary S1, $\pi(y) > 0$ iff $\ell(y) > 0$, a product set; any two of its points are joined by changing one coordinate at a time inside the set, each step with probability at least $\pi_i(\cdot \mid \cdot)/|F| > 0$, so the random-scan kernel $P = |F|^{-1} \sum_{i \in F} P_i$ is irreducible, and aperiodic since $P(y, y) > 0$. A finite, irreducible, aperiodic chain converges geometrically. The implementation sweeps the free points in a uniformly random order,

$P_{\mathrm{sw}} = \frac{1}{|F|!}\sum_\sigma P_{\sigma(1)}\cdots P_{\sigma(|F|)}$; each factor preserves π, and P_{sw} is reversible because the adjoint of a product is the product in reverse order and the set of permutations is closed under reversal. □

Proof of Theorem 1(b). Choose a bag uniformly; if it is not pure, choose uniformly an ordered pair (i, j) in it with $y_i \neq y_j$ and propose y' by swapping their labels. Brook's lemma through the intermediate state (y'_i, y_{-i}) gives

$$\frac{p(y' \mid X)}{p(y \mid X)} = \frac{p(y'_i \mid y_{-i}, X)}{p(y_i \mid y_{-i}, X)} \cdot \frac{p(y'_j \mid y'_i, y_{-ij}, X)}{p(y_j \mid y'_i, y_{-ij}, X)},$$

and Lemma 1 with (A1) turns the right-hand side into the ratio R_{ij} of four TFM predictives that the algorithm computes with two forward passes. A swap preserves the counts, so $\ell(y') = \ell(y)$ and $R_{ij} = \pi(y')/\pi(y)$, independently of the order of (i, j). The number of differently labelled pairs depends only on the counts, so the proposal is symmetric and the chain is Metropolis with stationary distribution π. Two assignments of a bag with the same counts are joined by a sequence of swaps preserving the counts, each with positive probability by Corollary S1; bags are independent, so the product chain is irreducible (the lazy version is also aperiodic). □

Full statement of Theorem 1.: (a) For instance-level weak labels the single-site chain satisfies detailed balance with respect to π and is irreducible and aperiodic on $\mathrm{supp}(\pi)$, hence converges geometrically. (b) For LLP the swap chain, whose acceptance ratio needs two forward passes, is reversible and irreducible on $\mathrm{supp}(\pi)$. (c) The inductive estimator (3) converges almost surely to $\bar{p}(y_* \mid x_*) = \sum_y \pi(y)\, p(y_* \mid x_*, (X, y))$, and $|\bar{p} - p(y_* \mid x_*, X, W)| \leq \mathrm{TV}\big(p(y \mid X, W),\, p(y \mid x_*, X, W)\big)$, which is zero when the prior is *separable*; the transductive chain converges to $p(y_* \mid x_*, X, W)$ exactly.

Proof of Theorem 1(c). (Inductive.) The ergodic theorem for a finite irreducible chain applied to the bounded function $y \mapsto p(y_* \mid x_*, (X, y))$, with (A1), gives almost sure convergence of (3) to $\bar{p}$. By the law of total probability and (A0), $p(y_* \mid x_*, X, W) = \sum_y p(y \mid x_*, X, W)\, p(y_* \mid x_*, (X, y))$; the difference of the expectations of one $[0, 1]$-valued function under two measures is at most their TV distance. If the prior is separable, $\phi = (\phi_x, \phi_{y|x})$ independent a priori with $p(x \mid \phi) = p(x \mid \phi_x)$ and $p(y \mid x, \phi) = p(y \mid x, \phi_{y|x})$, then $p(y \mid x_*, X, W) \propto \ell(y) \int p(y \mid X, \phi_{y|x}) p(\phi_{y|x}) d\phi_{y|x}$ times a factor that does not depend on y, so $p(y \mid x_*, X, W) = p(y \mid X, W)$ and the bound is zero. (Transductive.) Adding x_* as a free point with $\ell_* \equiv 1$ gives a chain for $p(y, y_* \mid X, x_*, W)$ that is exact by part (a) (for LLP, the swap chain combined with single-site updates of x_*, each preserving the target). By Lemma 1, the Rao–Blackwell term is $\tilde{p}(y_* \mid x_*, (X, y^{(s)}))$, whose average converges to $p(y_* \mid x_*, X, W)$. □

Proposition S1 (Bayes optimality under the prior)**.** *For any strictly proper scoring rule S, $p(y_* \mid x_*, X, W)$ uniquely minimises the Bayes risk $\mathbb{E}[S(f, y_*)]$ over all measurable predictors f of (X, W, x_*), the expectation being taken under the TFM's prior.*

Proof of Proposition S1. By the tower property, $\mathbb{E}[S(f, y_*)] = \mathbb{E}\big[\mathbb{E}[S(f, y_*) \mid X, W, x_*]\big]$. For each value of (X, W, x_*), f is a fixed distribution and $y_* \sim p(\cdot \mid x_*, X, W)$; strict propriety gives the unique minimiser $f = p(\cdot \mid x_*, X, W)$. Its risk is finite (at most $\log C$ for the log score). □

A.3 Theorem 2

For the random-scan single-site kernel, with $\varepsilon_{\mathrm{TV}}$ the TV error of the TFM's conditionals and κ a constant that depends only on ℓ, $\bar{\varepsilon} \leq \min\{1, 2\kappa\varepsilon_{\mathrm{TV}}\}$, and predictions inherit the error additively; both are shown at the end of the proof.

Proof of Theorem 2. Existence and uniqueness. The state space is finite, so $\tilde{P}$ has a stationary distribution. With $\tilde{p} > 0$ every single-site update and every MH acceptance probability is positive, so $\tilde{P}$ has the transition graph of P and is irreducible; $\tilde{\pi}$ is unique. Compatibility of the conditionals of $\tilde{p}$ is never used. *Perturbation identity.* Let $\Delta = \tilde{\pi} - \pi$ and $\nu = \tilde{\pi}(\tilde{P} - P)$, so $\nu(\Omega) = 0$. From $\tilde{\pi} = \tilde{\pi}\tilde{P}$ and $\pi = \pi P$, $\Delta = \nu + \Delta P$, and iterating, $\Delta = \sum_{t<T} \nu P^t + \Delta P^T$. *Contraction.* For a signed measure μ with $\mu(\Omega) = 0$, $\|\mu K\|_1 \leq \vartheta(K)\|\mu\|_1$. If $\tau(P) < \infty$ then $\vartheta(P^T) \to 0$, the remainder vanishes, and $\|\Delta\|_1 \leq \|\nu\|_1 \tau(P)$; if $\tau(P) = \infty$ there is nothing to prove (no aperiodicity is needed). Finally $\|\nu\|_1 \leq \sum_y \tilde{\pi}(y) \|\tilde{P}(y, \cdot) - P(y, \cdot)\|_1 = 2\bar{\varepsilon}$. *Single-site error.* $\tilde{P}(y, \cdot) - P(y, \cdot)$ averages the differences of the single-site updates, and each is the TV distance between $\tilde{p}\ell_i$ and $p\ell_i$ after normalisation. With $Z_p = \sum_k p(k)\ell_i(k)$, $2\,\mathrm{TV}(\tilde{p}_\ell, p_\ell) \leq \big(\sum_k \ell_i(k)|\tilde{p}(k) - p(k)| + |Z_q - Z_p|\big)/Z_p \leq 4(\max_k \ell_i(k)/Z_p)\,\mathrm{TV}(\tilde{p}, p)$, so the error is at most $2\kappa\varepsilon_{\mathrm{TV}}$ with $\kappa = \sup \max_k \ell_i(k)/Z_p$. *Prediction.* $|\mathbb{E}_{\tilde{\pi}}\tilde{p} - \mathbb{E}_\pi p| \leq |\mathbb{E}_{\tilde{\pi}}(\tilde{p} - p)| + |\mathbb{E}_{\tilde{\pi}} p - \mathbb{E}_\pi p| \leq \varepsilon'_{\mathrm{TV}} + \mathrm{TV}(\tilde{\pi}, \pi)$. □

Relation to classical bounds. With $E = \tilde{P} - P$ and the fundamental matrix Z, $\tilde{\pi} - \pi = \nu Z$ [2, 3], which gives $\mathrm{TV}(\tilde{\pi}, \pi) \leq \tau_1(Z)\bar{\varepsilon}$ with $\tau_1(Z) \leq \tau(P)$ [4, 5]; the bound of Theorem 2 is the weaker, more interpretable form. It is attained by a two-state chain. For a single-site sweep the per-sweep error is at most $|F|$ times the per-update error; the MH swap kernel has error at most $4\varepsilon_\infty$, with $\varepsilon_\infty = \sup|\log\tilde{p} - \log p|$.

A.4 Block updates

Proposition S2 (Blocks)**.** *(a) A block update whose proposal is the product of block-wise TFM predictives (for LLP, conditioned on bag counts), accepted with the MH ratio computed by Brook's lemma along the changed labels, leaves π invariant when $\tilde{p} = p$. (b) Without the MH step, Theorem 2 applies with an error $\bar{\varepsilon}_B$ that is bounded only under an extra stability assumption on the PPD, and the bound grows with n for fixed K.*

Proof of Proposition S2. (a) The proposal Q_B depends only on y_{-B}, which the step does not change, so the update is MH-within-Gibbs with an independent proposal. Brook's lemma along the changed labels with Lemma 1 gives the exact ratio $p(y' \mid X)/p(y \mid X)$, multiplied by the ratio of ℓ (one for LLP). The proposal is positive on the support of the block and acceptance is positive, so the chain is irreducible. (b) Let P_{bG} be exact block Gibbs. Under the stability assumption that adding one labelled point or one unlabelled feature to a context changes the PPD by at most β in TV, the chain rule and a hybrid argument give $\mathrm{TV}\big(\prod_{i\in B\cap F} p(\cdot \mid x_i, D_{-B}),\, p(y_{B\cap F} \mid y_{-(B\cap F)}, X)\big) \le m_f(m-1)\beta$ for a block of m points of which m_f are free, and after reweighting by ℓ, $\bar{\varepsilon}_B \le 2\kappa_B\, m_f(m-1)\beta$; Theorem 2 with $P = P_{\mathrm{bG}}$ concludes. □

Why this is vacuous at our sizes. With $\beta = O(1/n)$ and fixed K ($m = n/K$), $m_f(m-1)\beta = O(n/K^2)$ grows with n. For bag-blocks, κ_B is the inverse probability that all bags of a group match their counts under the product proposal: for $n \approx 2{,}000$ and $K = 10$, $\log_{10}\kappa_B \approx 15.8$ for bags of 8. On an exactly solvable LLP problem (3 bags of 4), the uncorrected bag-block chain has $\mathrm{TV}(\tilde{\pi}, \pi)$ between 0 and about 0.6 depending on the prior and grouping, so the approximation is real at small n. Section A.5 explains why bounds of this kind cannot be rescued and what can be said about predictions instead.

A.5 Uncorrected blocks: bias of the predictor

Proposition S2(b) and Theorem 2 bound the distance between label distributions. For predictions a first-moment argument is sharper. We state it in a pooled conjugate model, in which it is exact.

Model.: C classes with prior $\mathrm{Dir}(\alpha, \dots, \alpha)$ and no features. Of n points, N are labelled, L_c of them in class c, and $U = n - N$ are free with no weak label (semi-supervised learning). The state is the vector $k = (k_c)$ of class counts among the free points. The exact PPD of a new point is $f_c(k) = (L_c + k_c + \alpha)/(n + C\alpha)$, so the stationary mean of the predictor (3) is the stationary mean of f_c. One step redraws the free points of a random block given all labels outside the block: the exact kernel P draws them jointly from their conditional distribution (so the posterior π is stationary), the uncorrected kernel $\tilde{P}$ independently from the PPD given the context, as Algorithm 1 does. Blocks are either *free* (a uniform subset of u free points) or *all* (a uniform subset of $M = n/K$ of all points, the default of our sampler: the context then also loses the labelled points of the block).

Proposition S3 (Bias of the predictor under uncorrected blocks)**.** *(a) Let $P, \tilde{P}$ be Markov kernels on a finite state space and f a function with $\mathbb{E}_P[f(Y') \mid Y = y] - f(y) = A - a\, f(y)$ for all y, with $a \neq 0$. Let $d(y) = \mathbb{E}_{\tilde{P}}[f(Y') \mid y] - \mathbb{E}_P[f(Y') \mid y]$. Then for any stationary distributions π of P and $\tilde{\pi}$ of $\tilde{P}$, $\tilde{\pi}(f) - \pi(f) = \tilde{\pi}(d)/a$. (b) In the model above with free blocks, $\tilde{\pi}(f_c) = \pi(f_c)$ for all n, $u \le U$ and c. (c) With blocks of $M < n$ points over all points, $\mathbb{E}_{\tilde{\pi}}[k_c] = U(L_c + \alpha')/(N + C\alpha')$ with $\alpha' = \alpha(n-1)/(n-M)$, which is $\alpha\, K(n-1)/((K-1)n)$ for $M = n/K$: the counts of the uncorrected chain have exactly the posterior mean of the model with prior $\mathrm{Dir}(\alpha', \dots, \alpha')$. Consequently, for $M = n/K$,*

$$\tilde{\pi}(f_c) - \pi(f_c) = \frac{U(\alpha' - \alpha)(N - CL_c)}{(n + C\alpha)(N + C\alpha')(N + C\alpha)} = \frac{\alpha(1-\lambda)(1 - Cp_c)}{(K-1)\,N} \cdot \frac{n - K}{n + C\alpha} \cdot \frac{N^2}{(N + C\alpha')(N + C\alpha)},$$

with $\lambda = N/n$ and $p_c = L_c/N$; the last two factors are $1 - O(K/n) - O(C\alpha/N)$. The bias is governed by the number of labelled points: it does not vanish as n grows with N fixed. The predictor itself is not the posterior mean under α', only the counts are.

Proof. (a) Stationarity of $\tilde{\pi}$ gives $0 = \sum_y \tilde{\pi}(y)\big(\mathbb{E}_{\tilde{P}}[f(Y') \mid y] - f(y)\big) = \tilde{\pi}(A - af + d)$, so $\tilde{\pi}(f) = (A + \tilde{\pi}(d))/a$; the same with $d \equiv 0$ gives $\pi(f) = A/a$. (b) Let j_c be the number of free points of the block currently in class c. Given the block, the context has $n - u$ points with class counts $L_c + k_c - j_c$. Under P the new labels of the block are Dirichlet–multinomial, under $\tilde{P}$ independent categorical; both have marginals $(L_c + k_c - j_c + \alpha)/(n - u + C\alpha)$, hence the same expected new count, and $d \equiv 0$ for $f = k_c$. Averaging over the block ($\mathbb{E}[j_c] = uk_c/U$) gives the drift $\mathbb{E}_P[k_c' - k_c \mid k] = A - a\, k_c$ with $a = \frac{u}{U}\big(1 - \frac{U-u}{n-u+C\alpha}\big) > 0$, so (a) applies; f_c is affine in k_c. (c) Let l be the number of labelled points in the block and $u = M - l$; l is hypergeometric with $\mathbb{E}l = MN/n$ and $\mathrm{Var}\, l = V = M\frac{N}{n}\frac{U}{n}\frac{n-M}{n-1}$. Given l, the block contains in expectation lL_c/N labelled points of class c and uk_c/U free points currently in class c, and each of its u free points is redrawn from the PPD given the context, $(L_c - l_c + k_c - j_c + \alpha)/D$ with $D = n - M + C\alpha$. Hence

$$\mathbb{E}_{\tilde{P}}[k_c' - k_c \mid k] = \frac{L_c\, \mathbb{E}[u(1 - l/N)] + \alpha\, \mathbb{E}u}{D} - k_c\, \frac{\mathbb{E}u\,(D - U) + \mathbb{E}u^2}{DU}.$$

Using $\mathbb{E}u = MU/n$, $\mathbb{E}[ul] = M^2NU/n^2 - V$ and $\mathbb{E}u^2 = V + M^2U^2/n^2$, the two brackets simplify to $\frac{MU}{n}\big[L_c\frac{n-M}{n-1} + \alpha\big]$ and $\frac{MU}{n}\big[N\frac{n-M}{n-1} + C\alpha\big]$. Setting the drift to zero under $\tilde{\pi}$ gives $\mathbb{E}_{\tilde{\pi}}k_c = U\big[L_c + \alpha\frac{n-1}{n-M}\big]/\big[N + C\alpha\frac{n-1}{n-M}\big]$, and $\frac{n-1}{n-M} = \frac{K(n-1)}{(K-1)n}$ when $M = n/K$. Under π, by exchangeability, $\mathbb{E}_\pi k_c = U(L_c + \alpha)/(N + C\alpha)$. The displayed difference follows from $f_c(k) = (L_c + k_c + \alpha)/(n + C\alpha)$ and $\alpha' - \alpha = \alpha(n - K)/((K-1)n)$. □

Reading.: Part (a) is the prediction-level counterpart of Theorem 2: the bias is (expected error of one step) $\times$ (relaxation time of the mean, $1/a$), and it does not involve how far the joint distribution of a block is from the product of its marginals. In (c), removing the labelled points of the block from the context shrinks the chain towards the prior mean, by an amount proportional to $1/n$, to $1/(K-1)$, and to the ratio of free to labelled points; the bias vanishes when the labelled frequencies equal the prior mean. The closed form of (c) agrees with the fixed point of the affine drift to within 1.5×10^{-11} in 15 configurations, and the drift fixed points agree with the stationary means computed from full transition matrices to within 4×10^{-11} in 9 further configurations with $n \le 200$.

What the identity does not cover.: (i) The scan of Proposition S3 draws a fresh block at every step; Algorithm 1 partitions a permutation, for which the identity is not exact: we have only numerical agreement of the leading term. (ii) With weak labels that tilt the draws (PU, PLL), the drift of the exact kernel is no longer affine and free blocks are biased: in the pooled PU model the predictor is off by up to 0.070 with free blocks and by at most 0.004 with blocks over all points, which is why the latter are our default. (iii) For LLP the uncorrected bag-block kernel is biased already in its first moment. In a model with two feature types and identical bags (33 configurations), the predictor is off by up to 0.052 when $n \le 160$ and by at most 0.0024 from $n = 320$ on at $K = 10$, and by up to 0.13 at $K = 5$. (iv) A real TFM is not conjugate; the mechanism in (c) is still present, but its effective prior strength is unknown.

Why label-level bounds are vacuous.: In the pooled SSL and PU models with blocks over all points (24 configurations, $n \le 500$, $K \in \{5, 10, 20\}$), the exact distance $\mathrm{TV}(\tilde{\pi}, \pi)$ lies between 0.0012 and 0.0248 and does not decrease with n (for SSL at $K = 10$: 0.005, 0.009, 0.010, 0.011 at $n = 50, 100, 200, 500$), while the bound $\tau_1(Z)\bar{\varepsilon}$ exceeds 1 in 22 of them and is 60 to 629 times too large. The predictor, in contrast, is off by at most 0.0056. A guarantee for uncorrected blocks should therefore be stated for predictions, as in Proposition S3, and not for label vectors.

A.6 Dynamics of the sampler as run

Setting.: One sweep of Algorithm 1 draws a uniform permutation of $[n]$, independent of the past, and cuts it into K consecutive blocks $B_1, \dots, B_K$ of at most $m = \lceil n/K \rceil$ points. For $k = 1, \dots, K$ the free points of B_k are redrawn independently, each from the TFM's predictive given the current labels of all points outside B_k, multiplied by its weak-label factor. Every free point is redrawn exactly once per sweep. The TFM is a deterministic function of (context, query); the weak-label factors are fixed (for PU, known parameters); the context is not subsampled ($n - m$ does not exceed the context limit). We write $f_k = |B_k \cap F|$, $r_k = r_{-B_k}(Y_{s,k-1})$ for the accuracy of the context when block k is updated, $U_k = F \cap (B_1 \cup \cdots \cup B_{k-1})$ and $V_k = F \cap (B_{k+1} \cup \cdots \cup B_K)$ for the free points already and not yet updated, o_i for the correctness of point i at the start of the sweep and ξ_i for the correctness of its redraw. $\mathcal{F}_s$ is the history up to the end of sweep s. Since $n_F > m$, $F \setminus B_k \neq \emptyset$ and r_k is well defined. We use the conventions $\inf \emptyset = +\infty$ and $\sup \emptyset = -\infty$ in $\underline{\Phi}$ and $\overline{\Phi}$.

Lemma S1 (One sweep). *(i) Assume* (H_δ^-) *and let* x *be an* $\mathcal{F}_s$*-measurable level with* $\underline{\Phi}(x - \nu) - \delta \ge x$*. On the event* $\{R_s \ge x\}$*,* $P\big(R_{s+1} \ge \underline{\Phi}(x - \nu) - \delta - \lambda \mid \mathcal{F}_s\big) \ge 1 - p(\nu, \lambda)$*. (ii) Assume* (H_δ^+) *and let* x *be an* $\mathcal{F}_s$*-measurable level with* $\overline{\Phi}(x + \nu) + \delta \le x$*. On* $\{R_s \le x\}$*,* $P\big(R_{s+1} \le \overline{\Phi}(x + \nu) + \delta + \lambda \mid \mathcal{F}_s\big) \ge 1 - p(\nu, \lambda)$*.*

Proof. (i) Since $|U_k| + |V_k| = n_F - f_k$,

$$(n_F - f_k)\, r_k = \sum_{i \in U_k} \xi_i + \sum_{i \in V_k} o_i. \tag{S1}$$

Let $Z_k = \sum_{i \in U_k} (\xi_i - c_i)$, where c_i is the conditional probability that the redraw of i is correct given the state just before its block is updated, and let $\Xi_k = \sum_{i \in U_k} h_i + Z_k + \sum_{i \in V_k} (o_i - R_s)$ and $\mathcal{E} = \{\Xi_k > -\nu(n_F - m) \text{ for all } k\}$.

On $\mathcal{E}$*,* $r_k \ge x - \nu$ *for all* k*.* Induction on k. Suppose $r_j \ge x - \nu$ for $j < k$ (nothing to assume for $k = 1$). Then $\Phi(r_j) \ge \underline{\Phi}(x - \nu)$, and (H_δ^-) applied to each block $j < k$ with $f_j > 0$ gives $\sum_{i \in U_k} c_i \ge \sum_{i \in U_k} h_i + \sum_{j<k} f_j(\Phi(r_j) - \delta) \ge \sum_{i \in U_k} h_i + |U_k| x$. Also $\sum_{i \in V_k} o_i = |V_k| R_s + \sum_{i \in V_k} (o_i - R_s) \ge |V_k| x + \sum_{i \in V_k} (o_i - R_s)$. Adding the two in (S1), $(n_F - f_k) r_k \ge (n_F - f_k) x + \Xi_k$, and on $\mathcal{E}$, since $f_k \le m$, $r_k > x - \nu(n_F - m)/(n_F - f_k) \ge x - \nu$.

End of the sweep. Every free point is redrawn once, so $R_{s+1} = n_F^{-1}\big(\sum_{i \in F} c_i + Z\big)$ with $Z = \sum_{i \in F} (\xi_i - c_i)$. On $\mathcal{E}$, by (H_δ^-), $\sum_F h_i = 0$ and $\sum_k f_k = n_F$, $\sum_{i \in F} c_i \ge \sum_k f_k(\Phi(r_k) - \delta) \ge n_F(\underline{\Phi}(x - \nu) - \delta)$, hence $R_{s+1} \ge \underline{\Phi}(x - \nu) - \delta + Z/n_F$.

Probabilities. Give each of the three parts of Ξ_k the budget $t = \nu(n_F - m)/3 = \nu a n_F/3$. The set U_k is the intersection of F with the first $(k-1)$ blocks of the permutation, which is independent of $\mathcal{F}_s$; so $\sum_{U_k} h_i$ is a sum over a sample without replacement of at most n elements of the population $\{h_i \mathbb{1}[i \in F]\}_{i \le n}$, which has sum 0 and range at most $2h_{\max}$, and by Hoeffding's inequality for sampling without replacement [6, Section 6] it is below $-t$ with probability at most $\exp(-t^2/(2nh_{\max}^2))$. In the same way $\sum_{V_k} (o_i - R_s)$ is a sample from $\{(o_i - R_s)\mathbb{1}[i \in F]\}$, with sum 0 and range at most 1, and is below $-t$ with probability at most $\exp(-2t^2/n)$. Conditionally on the permutation, Z_k is a martingale in the order of the redraws, with at most n_F increments; each increment $\xi_i - c_i$ lies in the interval $[-c_i, 1 - c_i]$ of length 1, whose endpoints are determined by the past, so Hoeffding's lemma applied conditionally to each increment (the martingale form of Hoeffding's inequality with predictable ranges) gives $P(Z_k \le -t) \le \exp(-2t^2/n_F)$. With $n = n_F/\rho$ and $\rho \le 1$, each of the three is at most $\exp(-c\rho n_F \nu^2)$, and a union bound over $k \le K$ gives $P(\mathcal{E}^c \mid \mathcal{F}_s) \le 3Ke^{-c\rho n_F \nu^2}$. Finally $P(Z/n_F \le -\lambda) \le e^{-2n_F\lambda^2}$ by the same inequality. (If $h_{\max} = 0$ the first part is zero and $c = 2a^2/9$.) Part (ii) is symmetric, with $\overline{\Phi}$, the upper half of the assumption and the opposite tails. □

Proof of Theorem 3. Let $\psi(x) = \underline{\Phi}(x-\nu)-\delta-\lambda$, which is non-decreasing. (a) The condition $\psi(x) \geq x$ implies $\underline{\Phi}(x-\nu)-\delta \geq x$, so by Lemma S1(i), on $\{R_{s'} \geq x\}$, $R_{s'+1} \geq \psi(x) \geq x$ with conditional probability at least $1-p$. Applying this at $s' = s, \ldots, s+H-1$ gives $P(R_{s'} \geq x \text{ for all } s < s' \leq s+H) \geq \max\{0, 1-p\}^H \geq 1 - Hp$ for every integer $H \geq 1$. If $p < 1$, $\mathbb{E}[\tau - s] = \sum_{H \geq 0} P(\tau - s > H) \geq \sum_{H \geq 0}(1-p)^H = 1/p$ (τ may be infinite); if $p \geq 1$ the claim is trivial, since $\tau - s \geq 1$. (b) is symmetric, with Lemma S1(ii). (c) Let $q_0 = r_0$, $q_{j+1} = \min\{\psi(q_j), b\}$, and let J be the first index with $q_J \geq b$; since $\psi(q_j) \geq q_j + \gamma$ while $q_j < b$, $J \leq \lceil (b - r_0)/\gamma \rceil$. If $R_j \geq q_j$ and $q_j < b$, then $\underline{\Phi}(q_j - \nu) - \delta = \psi(q_j) + \lambda \geq q_j$, so Lemma S1(i) with $x = q_j$ gives $R_{j+1} \geq \psi(q_j) \geq q_{j+1}$ with conditional probability at least $1-p$. By induction $P(R_j \geq q_j \text{ for all } j \leq J) \geq 1 - Jp$. □

Remarks.: (i) *What each conclusion needs.* Staying up needs only the lower half of the assumption, staying down only the upper half. The proofs use the assumption only at the pairs (state before a block update, block) of the sweeps in question, and for a floor only when the context accuracy r_{-B} is at least $x - \nu$; if a block has a single free point, the assumption constrains that point individually. (ii) *Two basins.* Suppose Φ is non-decreasing and Λ-Lipschitz and g changes sign from negative to positive at u. Any $x > u$ with $g(x) \geq \delta + \Lambda\nu + \lambda$ is a floor; choosing $\nu = \lambda = (g(x) - \delta)/(\Lambda + 1)$ gives $p \leq (1 + 3K)\exp\big(-c\rho n_F (g(x) - \delta)^2/(\Lambda + 1)^2\big)$, so the expected time to fall below x grows exponentially with n_F; symmetrically below u. (iii) *Looseness.* The margin ν pays for the worst case in which all blocks of a sweep see a context that is off in the same direction, and Hoeffding's inequality uses ranges instead of variances. In simulations with a synthetic TFM whose response map is known and for which the assumption holds with $\delta = 0$ ($n = 3{,}000$, $K = 10$, per-point offsets $h_{\max} = 0.02$, the map $\Phi(r) = 0.45 + 0.5r$ with fixed point $r^* = 0.9$), the proven invariant band for $H = 60$ sweeps at total failure probability 0.05 reaches 0.22 below r^* (and is cut at 1 above), while the simulated 99% band has half-width 0.02. For a slope-one map with $\delta = 0$, $K = 10$, $h_{\max} \leq \frac{1}{2}$ and all labels unknown, a floor over 30 sweeps at failure probability 0.05 needs a gain margin $\nu + \lambda$ of 0.30, 0.17, 0.10 and 0.03 at $n_F = 10^3$, 3×10^3, 10^4 and 10^5. The qualitative statements hold for every n, but the bands are informative only for large n_F. (iv) *The size of δ for real TFMs.* If Assumption 1 holds, two states with the same r give responses, averaged over a sweep, within 2δ of each other (up to the small difference between r_{-B} and r). The response maps of Appendix G measured with chain-seeded and with uniform errors at equal r therefore give an estimated lower bound on δ: on the 78 PLL cells within the theorem's scope (16 datasets, TabICL v2 and TabPFN v2), $\delta \geq 0.05$ in the median cell, 0.12 at the 90th percentile and 0.14 at most, while uniform and structured placements differ by only 0.009 (median of the same bound). These are exploratory estimates that include measurement noise. (v) *Subsampled contexts.* When the context is subsampled to a fixed size, the accuracy the TFM sees is that of a random subsample, which adds a deviation term to ν; we do not state that extension, and the empirical check of Appendix G includes cells with subsampled contexts. (vi) *LLP.* Bag updates draw the labels of a bag jointly given its counts, so the redraws within a block are not independent and the lemma does not apply.

Proposition S4 (Stationary moments for an affine map)**.** *In the setting of Lemma S1, let all points be free ($\rho = 1$), K divide n, and suppose that every redraw in block k is correct, independently given the context, with probability exactly $c_0 + b\, r_k$, where $0 < c_0$, $c_0 + b < 1$ and $0 \leq b < 1$ (a point-wise affine response, stronger than Assumption 1 with $\Phi(r) = c_0 + br$, $h \equiv 0$, $\delta = 0$, which constrains block averages only). Then (a) $\mathbb{E}[R_{s+1} \mid \mathcal{F}_s] = \alpha_0 + \alpha_1 R_s$ and $\mathrm{Var}[R_{s+1} \mid \mathcal{F}_s] = q_0 + q_1 R_s + q_2 R_s^2$, with coefficients given by a K-step recursion; (b) under the stationary distribution of the chain, $\mathbb{E}R = r^* = c_0/(1-b)$, the lag-one autocorrelation of (R_s) is α_1, $|\alpha_1| < 1$, and $\mathrm{Var}\, R \cdot (1 - \alpha_1^2 - q_2) = q_0 + q_1 r^* + q_2 r^{*2}$.*

Proof. Let $S = nR_s$, O_j the number of correct labels in block j at the start of the sweep, and N_k the number of correct redraws in block k. Because the partition is uniform, $(O_1, \ldots, O_K)$ is multivariate hypergeometric: its mean is affine in S and its covariance matrix is quadratic in S. Since $r_k = \big(\sum_{j<k} N_j + \sum_{j>k} O_j\big)/(n-m)$, we have $N_k = mc_0 + g_k^\top W_k + \varepsilon_k$, where $W_k = (N_{<k}, O)$, g_k is a constant vector and $\varepsilon_k = N_k - m(c_0 + br_k)$ has conditional mean zero given the past of the sweep and conditional variance $m(c_0 + br_k)(1 - c_0 - br_k)$, a quadratic function of W_k whose expectation needs only the mean and covariance of W_k. By induction on k, the mean of $(N_{\leq k}, O)$ is affine in S and its covariance quadratic in S; $R_{s+1} = \sum_k N_k/n$ inherits both, which is (a). For (b), the chain is finite, irreducible and aperiodic because every redraw is correct with probability in $(0,1)$, so it has a unique stationary distribution, under which $\mathbb{E}R = \alpha_0 + \alpha_1 \mathbb{E}R$. If $\alpha_1 = 1$ then $\alpha_0 = 0$ and $\mathbb{E}[R_{s+1} \mid R_s = 0] = 0$, impossible since $c_0 > 0$; so $\mathbb{E}R = \alpha_0/(1-\alpha_1)$. The recursion for the means is affine in S and can be evaluated at any real S; at $S = nr^*$ it gives $\mathbb{E}r_k = r^*$ and $\mathbb{E}N_k = m(c_0 + br^*) = mr^*$ for every k, so r^* is the fixed point of $R \mapsto \alpha_0 + \alpha_1 R$. The covariance of R_{s+1} and R_s is $\alpha_1 \mathrm{Var}\, R$. The law of total variance gives $\mathrm{Var}\, R = \mathbb{E}[\mathrm{Var}(R_{s+1} \mid R_s)] + \alpha_1^2 \mathrm{Var}\, R$; the first term is positive, so $\alpha_1^2 < 1$, and expanding it with $\mathbb{E}R^2 = \mathrm{Var}\, R + r^{*2}$ gives the last identity. □

The exact standard deviation agrees with the simulated one within a factor 0.985–1.011 (Table S1); the Jacobi approximation is 18%–21% too small, because the noise of early blocks propagates to later blocks through the context. The Gaussian band is an approximation; the proposition gives moments, not tails.

A.7 When the naive TFM suffices

Let $\tilde{\eta}_k(x) = P(\tilde{w} = k \mid x)$ be the function that a consistent learner fitted on naive labels $\tilde{w}$ approaches, and $\eta_k(x) = p(y = k \mid x)$ the true posterior.

TABLE S1
Proposition S4 against simulation with the actual sampler and a synthetic TFM with a known affine response map (PLL, $C = 10$, $q = 0.6$, $K = 10$; per-point offsets with $h_{\max} = 0.02$, which the proposition ignores). "Jacobi": the approximation that treats all blocks of a sweep as seeing the same context.

$\Phi(r)$	n	s.d. (exact)	s.d. (simulated)	ratio	s.d. (Jacobi)	99% band (Gaussian)	99% band (simulated)
$0.3 + 0.5r$	500	0.0346	0.0341	0.985	0.0274	[0.511, 0.689]	[0.512, 0.686]
$0.3 + 0.5r$	3,000	0.0142	0.0143	1.011	0.0112	[0.564, 0.636]	[0.563, 0.636]
$0.45 + 0.5r$	500	0.0190	0.0191	1.005	0.0155	[0.851, 0.949]	[0.844, 0.944]
$0.45 + 0.5r$	3,000	0.0078	0.0077	0.996	0.0063	[0.880, 0.920]	[0.880, 0.919]

Proposition S5 (When the naive TFM suffices)**.** *(a) If the naive label depends on x only through y, through a channel M with equal off-diagonal entries in each column ($M_{y,k} = b_k$ for $y \neq k$) and $M_{k,k} > b_k$, then $\tilde{\eta}_k = b_k + \lambda_k \eta_k$ with $\lambda_k = M_{k,k} - b_k > 0$, so the naive target ranks points exactly as η_k; this covers PU-SCAR, binary class-dependent noise with $T_{00} + T_{11} > 1$, SSL with labels missing completely at random, PLL in which every wrong class enters the candidate set independently with the same probability $q < 1$, and LLP with random bags. For accuracy with several classes, the arg max is preserved when b_k and λ_k do not depend on k, as for PLL with uniform distractors. (b) If the naive labels are drawn from the observation by a kernel that does not depend on the parameter, their Fisher information is at most that of the observation, with equality iff the observation's score is a function of the naive labels. Equality holds for PU-SCAR and label noise, where the naive labels are the observation, and for LLP when the naive labels match the bag counts. For PLL the inequality was strict in every case we computed. For LLP the loss arises at the learner: the point-wise (composite) likelihood on naive labels has a Godambe information of about 3% of the information in the counts for bags of 8 in a logistic model.*

Complement to Proposition S5(a).: For binary labels with class prior ϖ, for any score s, $\mathrm{AUC}(\eta) - \mathrm{AUC}(s) \leq \mathbb{E}|s - \tilde{\eta}|/(\lambda\varpi(1-\varpi))$: the AUC regret of a learner fitted to naive labels is its own estimation error, amplified by $1/\lambda$.

Proof of Proposition S5 and of the complement. (a) With $P(\tilde{w} = k \mid x, y) = M_{y,k}$ independent of x and $M_{y,k} = b_k$ for $y \neq k$, $\tilde{\eta}_k = M_{k,k}\eta_k + b_k(1 - \eta_k) = b_k + \lambda_k\eta_k$, $\lambda_k = M_{k,k} - b_k$. For PLL with each wrong class entering the candidate set independently with probability q and the naive label chosen uniformly from the set, $M_{y,y} = a = \mathbb{E}[1/(1+N)] = \frac{1-(1-q)^C}{Cq}$ with $N \sim \mathrm{Bin}(C-1, q)$ and $b = (1-a)/(C-1)$, so $\lambda = (Ca-1)/(C-1) > 0$ for $q < 1$. For LLP with random bags of size k, $P(\tilde{w}_i = 1 \mid x_i, y_i) = (y_i + (k-1)\varpi)/k$, so $\lambda = 1/k$. A strictly increasing affine map preserves ranks, hence one-vs-rest AUC. For the binary bound, write $\varpi(1-\varpi)\mathrm{AUC}(s) = \mathbb{E}[\eta(X)(1-\eta(X'))h_s(X, X')]$ for independent copies, with $h_s = \mathbb{1}[s(X) > s(X')] + \frac{1}{2}\mathbb{1}[s(X) = s(X')]$; symmetrising, $\varpi(1-\varpi)[\mathrm{AUC}(\eta) - \mathrm{AUC}(s)] = \frac{1}{2}\mathbb{E}\big[|\Delta|(\mathbb{1}[s \text{ reverses the pair}] + \frac{1}{2}\mathbb{1}[s \text{ ties}, \Delta \neq 0])\big]$ with $\Delta = \eta(X) - \eta(X')$. With $t = (s-b)/\lambda$, which has the ranks of s, a reversed or tied pair has $|\Delta| \leq |\eta(X) - t(X)| + |\eta(X') - t(X')|$, so the right-hand side is at most $\mathbb{E}|s - \tilde{\eta}|/\lambda$. For $\lambda = 1$ and scores that are class-probability estimates, this is Corollary 12 of Agarwal [7]. (b) $p_\theta(\tilde{w}) = \sum_O p_\theta(O)R(\tilde{w} \mid O)$ gives $\partial_\theta \log p_\theta(\tilde{w}) = \mathbb{E}_\theta[\partial_\theta \log p_\theta(O) \mid \tilde{w}]$, and Jensen's inequality for conditional expectations gives $I_{\tilde{w}}(\theta) \leq I_O(\theta)$, with equality iff the score of O is $\sigma(\tilde{w})$-measurable. □

Information lost by the naive labels (logistic model $\eta = \sigma(\theta x)$, $\theta = 2$, $x \sim \mathcal{N}(0, I)$; per point). LLP: the Godambe information of the pointwise (composite) likelihood on naive labels, relative to the information of the bag counts, is 0.134, 0.060, 0.033, 0.017, 0.009 for bags of 2, 4, 8, 16, 32. PLL: the naive labels retain 0.24–0.26 of the information in the candidate sets at $q = 0.3$ and 0.06–0.10 at $q = 0.6$ ($C \in \{3, 4, 7\}$). For PU-SCAR and label noise the ratio is 1.

APPENDIX B
PRE-REGISTRATION AND TIMELINE

All three studies were run under a written protocol with hypotheses, comparators, tests, datasets and decision rules fixed before any evaluation data were used; every later change was logged as a dated deviation. Each freeze is a commit, and every result file records the hashes of the code that produced it. A further supplementary archive lists the freeze events with the times of their commits and the SHA-256 of every frozen file, the scoring scripts, and SHA-256 hashes of the result files.

TABLE S2
Timeline of the three studies (2026, UTC). "Exploratory" marks analyses that were not part of the registered tests.

Date	Event
26 Oct	Study 1 protocol frozen (hypotheses H1, exactness on enumerable problems, already met before the freeze (Appendix I); H2, LLP; H3, against hard-EM on log loss; H4, PLL; H5, rank on real partial labels; datasets, baselines, tuning spaces, primary-comparator rule, tests, ablations, sensitivity analysis). Later changes were logged as dated deviations: all but three before the evaluation runs. The three made during the analysis are reported: dropping an external LLP benchmark whose data were unavailable; labelling the TFM ablation exploratory; and reading the 60% criterion of H4, which did not say whether it applies per level or pooled and to one strongest baseline or the best per dataset, in its weakest form. That reading changed no verdict, because the Wilcoxon test against tuned CC was not significant at either level.
28 Oct–1 Nov	Study 1 runs on the evaluation datasets (TabPFN v2; registered TabICL v2 ablation; baselines).
30 Oct	Study 1 analysis: H2 and H4 not met, H3 and H5 met; TFM ablation passes all tests (exploratory). Decision to run a confirmatory replication changing only the TFM.
1 Nov	Study 2 dataset census and selection from metadata only; flags for the sensitivity analysis set by hand.
1 Nov 15:19	**Study 2 frozen** (commit), before any Study 2 table was downloaded.
1–2 Nov	Data download and post-check rule applied (two datasets fail $n \geq 500$ after de-duplication; first reserve added, subsampled to 10,000 rows). Exploratory response-map runs on Study 1 datasets only.
2–3 Nov	Study 2 runs; confirmatory analysis. One post-hoc fix of a floating-point tie in the analysis (a tie counted as a loss); no conclusion changed. Failed cells of one baseline rerun once.
3 Nov	Exploratory runs on Study 1 datasets: $\widehat{R}$ and MH-corrected blocks with TabICL v2.
4–10 Nov	Development of the multi-start and multi-TFM samplers on earlier and development datasets (exploratory; Appendix G); analysis of the response-map dynamics (Theorem 3).
10 Nov 04:21	**Study 3 frozen** (commit), before its code, its scoring script and any download of its tables.
10–12 Nov	Study 3 runs (three sessions); one interim look at 60 of 71 pairs on 11 Nov; confirmatory scoring with the pre-written script on 12 Nov (03:18); exploratory decomposition afterwards (Appendix E).
12 Nov 07:21	**Supplementary analysis S3B frozen** (plan, scoring script, prediction P22; commit), with the results of Study 3 known; its code committed at 07:27, before any run.
12–13 Nov	S3B runs (one CPU and two GPU sessions); B1 and B2 scored; B3 scored on the complete TFM, one look at the other available cells, then scored again on all cells after the second session (Appendix F).

APPENDIX C
STUDY 1 WITH TABICL V2 (REGISTERED TFM ABLATION)

Study 1 in detail.: With TabPFN v2, In-Context Gibbs led the primary comparators on LLP and on PLL at $q = 0.3$, short of significance, and H2 and H4 were not met. On the four datasets with at least eight classes chains ended far from the truth (overall -0.132 against tuned CC at $q = 0.6$): the label accuracy stayed at its starting value on mfeat-factors (0.16–0.17), reached only 0.34–0.37 on semeion and 0.44–0.64 on MiceProtein, and on optdigits one seed of three stayed low (0.27, 0.89, 0.93; Table S4).

Table S3 applies the H2 and H4 tests of Study 1 to the registered TFM ablation, in which every TFM-based method uses TabICL v2. With clean labels the two TFMs are about equally accurate (oracle within 0.005); the difference appears only with weak labels. Because TabICL v2 was promoted after these results were seen, they are exploratory; Study 2 is the confirmatory test.

TABLE S3
Study 1 datasets, TabICL v2 for all TFM-based methods (exploratory). Same tests and comparators as Table 1.

Setting	Comparator	Δ [95% CI]	W/T/L	p_{Holm}	mean: In-Context Gibbs (TabICL / TabPFN)		/ oracle (TabICL)
LLP, bag 8	DLLP (linear)	+0.058 [+0.029, +0.093]	10/0/0	0.004	0.889	0.855	0.928
PLL, $q = 0.3$	CC, tuned	+0.029 [+0.013, +0.051]	12/0/2	0.004	0.854	0.847	0.865
PLL, $q = 0.6$	CC, tuned	+0.038 [+0.013, +0.071]	12/0/2	0.009	0.831	0.661	0.865

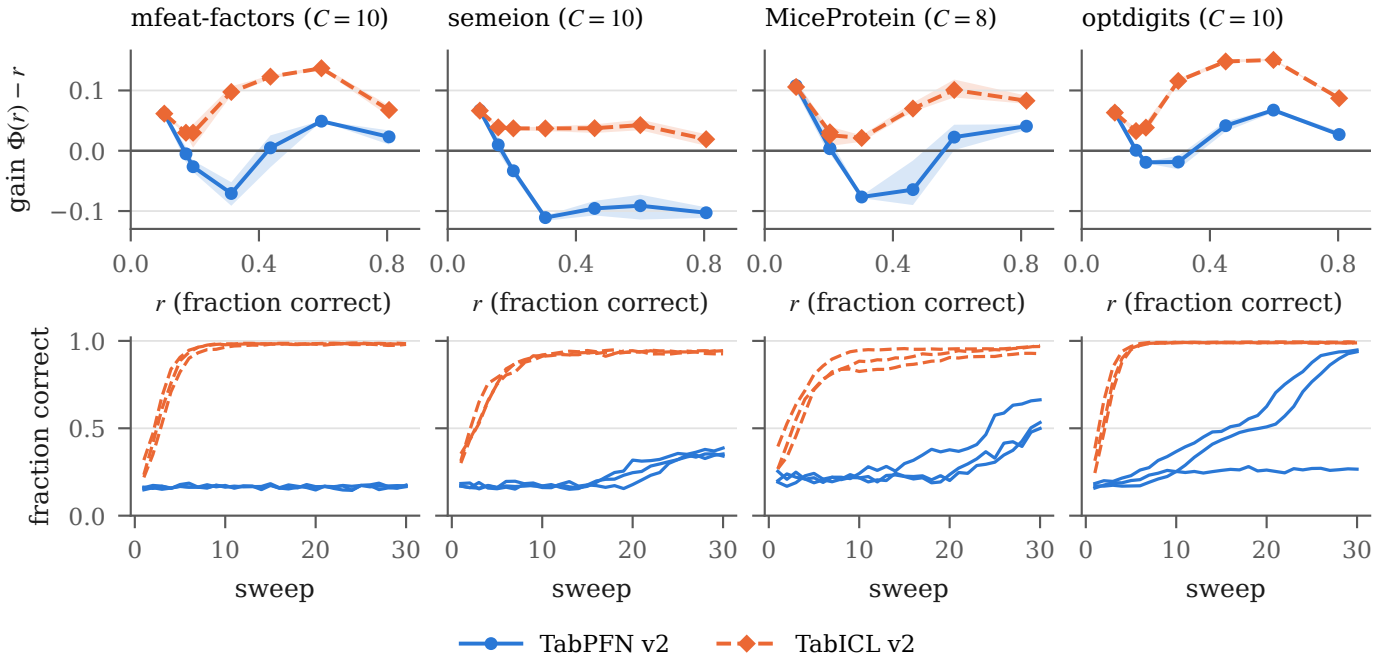


Fig. S1. PLL at $q = 0.6$ on the four Study 1 datasets with at least eight classes. Top: response maps as in Figure 1. Bottom: fraction of correct training labels along the chains of the main runs, one line per seed (diagnostic only; the sampler never sees the true labels). With TabPFN v2 several chains end far below the truth, though not all stay within the region of negative gain (Table S4); with TabICL v2 all chains reach high label accuracy within 5–15 sweeps.

TABLE S4
Label accuracy of the single chains of Study 1 (PLL, $q = 0.6$) on the four datasets with at least eight classes: mean value after the first sweep, final value (mean of the last five sweeps) and test accuracy for each of the three seeds, sorted. Label accuracies use the true training labels and are diagnostics only. Chains whose label accuracy rose by more than 0.03 between sweeps 16–20 and 26–30, i.e. were still rising when they stopped: TabPFN v2, semeion: 3 of 3; TabPFN v2, MiceProtein: 3 of 3; TabPFN v2, optdigits: 2 of 3; TabICL v2, MiceProtein: 2 of 3.

TFM	Dataset	after sweep 1	final label accuracy (3 seeds)	test accuracy (3 seeds)
TabPFN v2	mfeat-factors	0.16	0.16, 0.17, 0.17	0.10, 0.10, 0.10
TabPFN v2	semeion	0.17	0.34, 0.35, 0.37	0.27, 0.30, 0.31
TabPFN v2	MiceProtein	0.22	0.44, 0.44, 0.64	0.43, 0.44, 0.56
TabPFN v2	optdigits	0.17	0.27, 0.89, 0.93	0.29, 0.92, 0.96
TabICL v2	mfeat-factors	0.26	0.98, 0.98, 0.98	0.97, 0.98, 0.99
TabICL v2	semeion	0.32	0.93, 0.94, 0.94	0.95, 0.95, 0.96
TabICL v2	MiceProtein	0.31	0.93, 0.96, 0.96	0.92, 0.93, 0.96
TabICL v2	optdigits	0.31	0.99, 0.99, 0.99	0.99, 0.99, 1.00

APPENDIX D
STUDY 2: SENSITIVITY ANALYSIS, ALL BASELINES AND PER-DATASET RESULTS

Further results.: In accuracy the single chain was ahead of hard-EM at $q = 0.6$ (0.851 against 0.803) and level at $q = 0.3$ (0.862 against 0.862). Averaging over samples adds little to predicting from the last sample alone ("stochastic EM": 0.843; Table S9). The gain over the naive TFM is absent where naive labels carry all the information (Appendix M). In an exploratory rerun of Study 2 with a second chain, H4′ at $q = 0.6$ does not pass ($p_{\text{Holm}} = 0.074$), while with the two chains averaged every test passes (Table S26).

Sensitivity.: Five datasets were flagged before any data were downloaded (repeated measurements of the same units, or temporal drift): SpeedDating and eye_movements (binary); JapaneseVowels, cjs and gas-drift (multiclass). Without them, every conclusion is unchanged (Table S5). With seven multiclass datasets, Holm-adjusted Wilcoxon tests cannot reach 0.05 unless every comparison is won; the registered rule for this case checks that the mean difference keeps its sign and that In-Context Gibbs still wins on at least 60% of the remaining datasets, which holds at both levels.

Rounding in Table 1.: The two PLL rows of Study 2 against tuned CC are different results that agree to three decimals: +0.0125 [+0.0065, +0.0192] at $q = 0.3$ and +0.0121 [+0.0056, +0.0193] at $q = 0.6$.

All baselines.: Against each baseline separately (Holm over all baseline–level pairs of a setting), In-Context Gibbs is significantly better than all ten LLP baselines; on PLL every mean difference is positive and 7 of 16 pairs are significant after this stricter correction (Table S6). One baseline (default PRODEN) produced non-finite probabilities on 5 of 6 gas-drift cells,

TABLE S5
Study 2 without the flagged datasets.

Setting	Comparator	datasets	Δ (all datasets)	Δ (flagged removed)	W/T/L	p_{Holm}	Δ vs naive
LLP, bag 8	DLLP (linear)	16	+0.052	+0.051	15/0/1	<0.001	+0.124
PLL, $q = 0.3$	CC, tuned	7	+0.012	+0.016	6/1/0	0.062	+0.023
PLL, $q = 0.6$	CC, tuned	7	+0.012	+0.013	5/1/1	0.062	+0.168

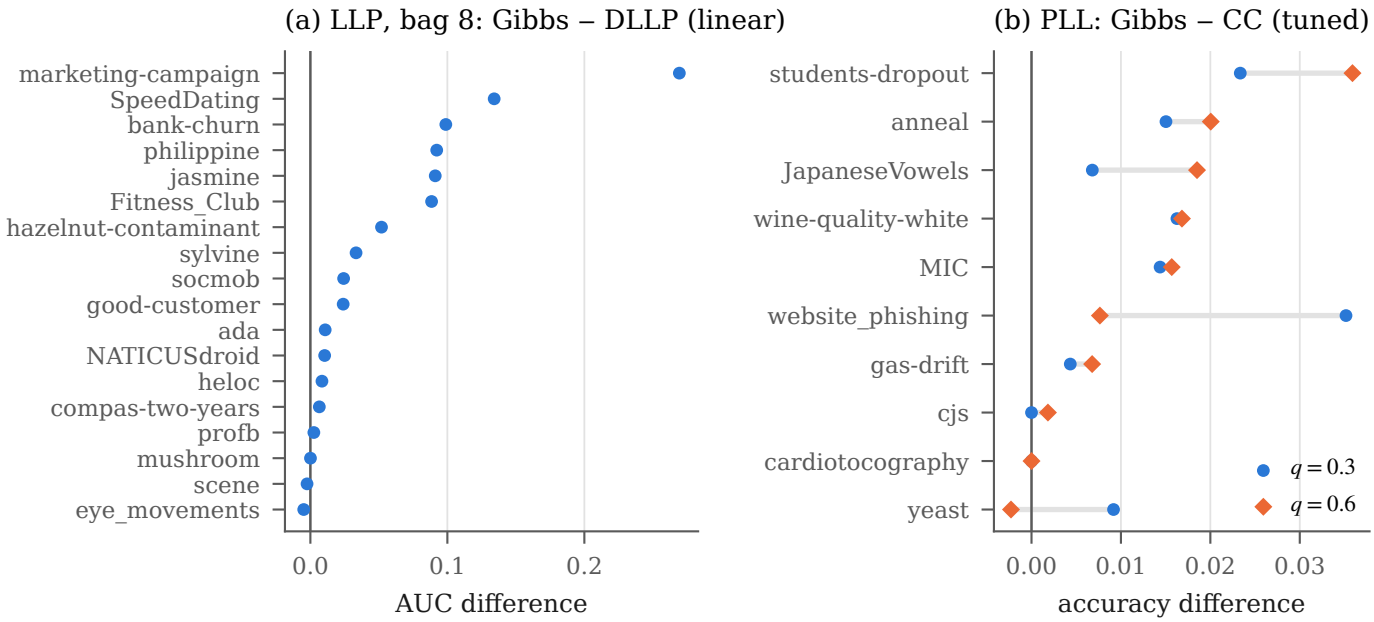


Fig. S2. Study 2, per dataset: In-Context Gibbs (single chain) minus the primary comparator (positive favours In-Context Gibbs), mean over 3 seeds. (a) LLP with bags of 8, AUC against linear DLLP. (b) PLL, accuracy against tuned CC at two candidate-set rates; a bar joins the two rates of a dataset.

also on a rerun; its means use the remaining cells.

TABLE S6
Study 2, In-Context Gibbs against every baseline. Bold: significant in favour of In-Context Gibbs.

Setting	Baseline	datasets	Δ	W/T/L	p_{Holm}
LLP, bag 8 (AUC)	DLLP (MLP)	18	+0.057	17/0/1	<**0.001**
	DLLP (MLP), tuned	18	+0.041	16/0/2	<**0.001**
	DLLP (linear)	18	+0.052	16/0/2	<**0.001**
	DLLP (linear), tuned	18	+0.046	16/0/2	<**0.001**
	LLP-VAT	18	+0.062	16/0/2	<**0.001**
	LLP-VAT, tuned	18	+0.042	16/0/2	<**0.001**
	EasyLLP	18	+0.116	18/0/0	<**0.001**
	EasyLLP, tuned	18	+0.069	17/0/1	<**0.001**
	GLWS	18	+0.056	17/1/0	<**0.001**
	GLWS, tuned	18	+0.051	18/0/0	<**0.001**
PLL, $q = 0.3$ (acc.)	PRODEN	9	+0.048	9/0/0	**0.043**
	PRODEN, tuned	10	+0.027	8/1/1	0.070
	GLWS/ILL	10	+0.048	10/0/0	**0.031**
	GLWS/ILL, tuned	10	+0.014	10/0/0	**0.031**
	RC	10	+0.034	9/0/1	**0.043**
	RC, tuned	10	+0.013	8/1/1	0.070
	CC	10	+0.035	10/0/0	**0.031**
	CC, tuned	10	+0.012	8/2/0	0.070
PLL, $q = 0.6$ (acc.)	PRODEN	10	+0.173	10/0/0	**0.031**
	PRODEN, tuned	10	+0.031	8/0/2	0.070
	GLWS/ILL	10	+0.128	10/0/0	**0.031**
	GLWS/ILL, tuned	10	+0.015	8/0/2	0.070
	RC	10	+0.048	8/0/2	0.070
	RC, tuned	10	+0.012	8/1/1	0.070
	CC	10	+0.047	8/0/2	0.070
	CC, tuned	10	+0.012	8/1/1	0.070

TABLE S7
Study 2, LLP with bags of 8: AUC per dataset (mean over 3 seeds). Bold: best non-oracle method among those shown.

Dataset	Gibbs	hard-EM	naive	DLLP-lin	DLLP tuned	LLP-VAT tuned	oracle
ada	**0.898**	0.895	0.742	0.888	0.880	0.883	0.912
Bank_Customer_Churn	0.849	**0.855**	0.785	0.750	0.837	0.834	0.877
compas-two-years	**0.683**	0.671	0.539	0.677	0.650	0.662	0.692
eye_movements	0.564	**0.589**	0.538	0.569	0.549	0.548	0.701
Fitness_Club	**0.831**	0.784	0.609	0.743	0.719	0.753	0.831
hazelnut-contaminant	**0.993**	0.989	0.853	0.941	0.953	0.951	0.995
heloc	**0.797**	0.788	0.759	0.789	0.788	0.787	0.813
Is-this-a-good-customer	0.633	0.634	0.576	0.610	**0.656**	0.631	0.737
jasmine	**0.870**	0.841	0.780	0.779	0.807	0.792	0.902
Marketing_Campaign	**0.878**	0.870	0.667	0.609	0.630	0.635	0.915
mushroom	**1.000**	**1.000**	0.721	1.000	0.996	1.000	1.000
NATICUSdroid	0.983	**0.985**	0.911	0.973	0.976	0.977	0.988
philippine	0.774	**0.781**	0.711	0.681	0.699	0.700	0.934
profb	0.580	0.495	0.507	0.578	0.569	**0.587**	0.675
scene	0.971	**0.983**	0.823	0.973	0.977	0.973	0.991
socmob	**0.980**	0.979	0.794	0.956	0.950	0.947	0.987
SpeedDating	**0.842**	0.752	0.768	0.708	0.783	0.745	0.871
sylvine	0.993	**0.996**	0.955	0.960	0.965	0.958	0.997
mean	0.840	0.827	0.724	0.788	0.799	0.798	0.879

TABLE S8
Study 2, PLL: accuracy per dataset (mean over 3 seeds). Bold: best non-oracle method among those shown.

	$q = 0.3$						$q = 0.6$					
Dataset	Gibbs	hard-EM	naive	CC tuned	RC tuned	oracle	Gibbs	hard-EM	naive	CC tuned	RC tuned	oracle
anneal	**0.991**	0.989	0.955	0.976	0.972	0.996	**0.977**	0.945	0.801	0.957	0.962	0.996
cardiotocography	**1.000**	**1.000**	0.994	**1.000**	**1.000**	1.000	**1.000**	0.886	0.476	**1.000**	**1.000**	1.000
cjs	0.997	0.997	0.962	0.997	**0.998**	1.000	**0.993**	0.830	0.391	0.991	0.989	1.000
gas-drift	0.996	**0.997**	0.993	0.991	0.992	0.998	**0.995**	0.994	0.930	0.988	0.986	0.998
JapaneseVowels	0.991	**0.995**	0.968	0.984	0.980	0.998	**0.989**	0.983	0.785	0.970	0.973	0.998
MIC	**0.887**	0.886	0.883	0.873	0.872	0.891	**0.886**	0.877	0.848	0.870	0.871	0.891
students-dropout	0.777	**0.782**	0.775	0.754	0.754	0.786	**0.770**	0.764	0.744	0.734	0.733	0.786
website_phishing	**0.823**	0.820	0.792	0.787	0.787	0.844	0.786	**0.787**	0.735	0.778	0.778	0.844
wine-quality-white	0.563	**0.571**	0.538	0.547	0.546	0.585	**0.556**	0.510	0.481	0.539	0.537	0.585
yeast	**0.597**	0.578	0.543	0.588	0.591	0.606	0.561	0.456	0.275	**0.563**	0.563	0.606
mean	0.862	0.862	0.840	0.850	0.849	0.870	0.851	0.803	0.647	0.839	0.839	0.870

TABLE S9
Study 2: mean score of every method (datasets averaged over 3 seeds). PRODEN (default) failed on gas-drift in 5 of 6 cells (non-finite probabilities, reproduced on a rerun); its $q = 0.3$ mean is over 9 datasets and its $q = 0.6$ gas-drift entry uses one seed.

LLP, bag 8 (AUC, 18 datasets)			PLL (accuracy, 10 datasets)				
Method	default	tuned	Method	$q = 0.3$		$q = 0.6$	
				default	tuned	default	tuned
In-Context Gibbs (ours)	0.840		**In-Context Gibbs (ours)**	0.862		0.851	
Hard-EM (same TFM)	0.827		Hard-EM (same TFM)	0.862		0.803	
Stochastic EM (same TFM)	0.835		Stochastic EM (same TFM)	0.858		0.843	
Naive TFM	0.724		Naive TFM	0.840		0.647	
DLLP (MLP)	0.783	0.799	PRODEN	0.800	0.836	0.678	0.820
DLLP (linear)	0.788	0.794	GLWS	0.814	0.848	0.724	0.837
LLP-VAT	0.778	0.798	RC	0.828	0.849	0.803	0.839
EasyLLP	0.724	0.771	CC	0.827	0.850	0.804	0.839
GLWS	0.784	0.789	Oracle (true labels)	0.870		0.870	
Oracle (true labels)	0.879						

APPENDIX E
STUDY 3: PROTOCOL, DATASETS AND FURTHER RESULTS

Protocol.: The protocol registers the hypotheses as H1–H3 and the criteria as C4 and C5; we write them with a double prime to keep them apart from those of Studies 1 and 2. It was committed before the code of the study, before the scoring script and before any Study 3 table was downloaded, and the scoring script was committed before any result existed. Nothing of the method was changed on these datasets: the chain settings are those of Section 4, the three starts and the equal weights were chosen on development data (Appendix G). A (TFM, dataset) pair counts if all three seeds ran without error; all 71 pairs did (213 cells). The protocol also recorded our expectations: that both hypotheses and all three criteria would hold. Two did not.

Datasets.: Candidates were taken, from metadata only, from the suites of Study 2 and OpenML-CC18, after excluding 176 dataset identifiers: every dataset used in Study 1, Study 2 or development, and the 123 datasets on which TabDPT was pretrained. The size and class-balance criteria of Study 2 were applied at their most relaxed registered tier (500 to 20,000 rows, at most 2,000 features, 3 to 20 classes, a minority class of at least 5% for binary tasks), which left 43 candidates; a manual review by Study 2's rules (real data only; no dataset from the same source as one already used; one dataset per source) kept 20 and removed 8 synthetic datasets, 14 that share a source with a dataset used earlier and 1 duplicate source. 6 datasets with more than 10,000 rows were subsampled to 10,000 (stratified, fixed seed) before splitting. 5 datasets were flagged in advance for group or sequence structure. Known weaknesses, recorded in the protocol: the multiclass group has only 8 datasets, which is why the tests pool both settings and per-setting results are descriptive; three datasets had been on the reserve list of Study 2 (their tables had been hashed, no method had been run on them); and we cannot tell whether the developers of TabPFN-3 or TabICL v2 used any of these datasets.

Execution.: Three sessions (10–12 November 2026). One change was made to the pre-written scoring script before any scoring: a download error at the dataset-hash step is treated as "not checked" instead of as a failed criterion (no dataset was affected). The order of the GPU queues was rebalanced between sessions. One interim look at 60 of the 71 pairs was taken on 11 November with the pre-written script; nothing was changed afterwards. The files of completed cells were byte-identical across sessions.

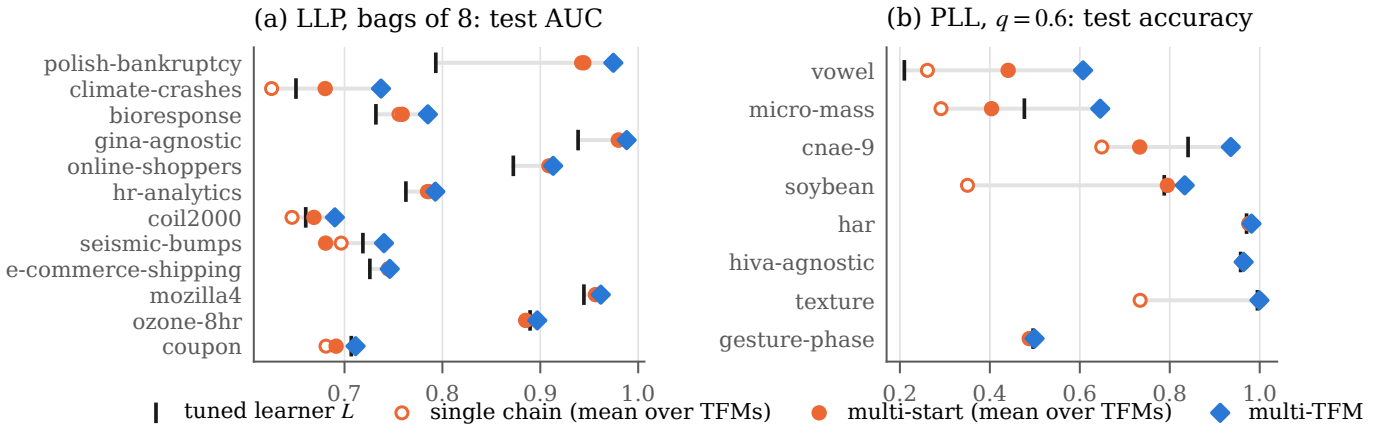


Fig. S3. Study 3, per dataset (mean over 3 seeds), ordered by the gain of the multi-TFM average over the tuned learner L. "Single chain" and "multi-start" are means over the TFMs run on the dataset.

TABLE S10
Study 3, mean test score over datasets (AUC for LLP, accuracy for PLL; number of datasets in brackets) of the single chain and the multi-start average of each TFM, of the multi-TFM average, and of the tuned learner L used in the learner start. †TabPFN v2 ran only on 10 LLP and 1 PLL datasets; its row is not comparable with the others.

	LLP, AUC (12)		PLL, acc. (8)	
TFM	single	multi	single	multi
TabICL v2	0.802	0.803	0.749	0.805
TabPFN-3	0.817	0.817	0.371	0.650
TabDPT	0.786	0.799	0.648	0.720
TabPFN v2†	0.784	0.798	0.488	0.494
Multi-TFM		**0.828**		**0.808**
Tuned learner L	0.783		0.717	

Registered criteria not met.: The diagnostic of H3″, the mean total-variation distance between the label marginals of the warm-start and learner-start chains, reached an AUC of 0.766 (66 failing cells of 213), below the registered 0.80. It separated failures for TabPFN-3 (0.93) and TabICL v2 (0.85) but not for TabDPT (0.59) or TabPFN v2 (0.64): we have no label-free diagnostic that transfers across TFMs. C4″ also failed: with one TFM, the multi-start sampler was more than 0.02 behind the better of its single chain and L in 10 of 71 pairs (14.1%): 2 behind the single chain, 9 behind L, 1 behind both (Table S13). The registered descriptive comparison of one TFM's multi-start average with L was not significant (+0.018, two-sided $p = 0.14$).

Regressions.: In 9 of the 10 pairs of Table S13 L is better than the multi-start average, by a wide margin where the single chain stalled deeply (cnae-9, micro-mass); in the remaining pair (TabPFN-3, seismic-bumps) the multi-start average is behind its own single chain only. The single chain itself was more than 0.02 below L in 14 of the 71 pairs. In 4 cells the single chain had an AUC below 0.5 on LLP (0.30–0.49).

Exploratory decomposition.: Table S14 separates the ingredients of the two averages. Within one TFM, everything that includes the learner start or the learner's prediction is close to the registered three-start average, and everything that does not is close to the single chain; adding L's prediction to the three-start average gives +0.021 ($p = 0.005$). Across TFMs, the average of the chains started from L equals the registered multi-TFM average (-0.001), the average of single chains differs from it by -0.028, and TabPFN v2, available on 11 datasets, makes no difference. With TabICL v2 alone, the configuration of Study 2, the single chain was above L by +0.019 on LLP (9/0/3) and +0.032 on PLL (7/0/1), and its multi-start average by +0.020 and +0.088 (10/0/2 and 8/0/0); here too single chains stalled on individual datasets (Table S11). These comparisons were chosen after seeing the confirmatory results and carry no error control.

Which configuration, on what evidence.: For a reader deciding what to run, Table S16 separates the three configurations by the kind of evidence behind them.

TABLE S11
Study 3, per dataset: test score of the single chain ("1") and the multi-start average ("ms") for each TFM, the multi-TFM average and the tuned learner L (mean over 3 seeds). n is the number of rows after de-duplication and subsampling. TabPFN v2 ran only within its published limits.

					TabICL v2		TabPFN-3		TabDPT		TabPFN v2			
ID	Dataset	n	d	C	1	ms	1	ms	1	ms	1	ms	multi-TFM	L
LLP, bags of 8 (AUC)														
1038	gina-agnostic	3,468	970	2	0.993	0.993	0.990	0.990	0.956	0.959	–	–	0.988	0.939
1046	mozilla4	10,000	5	2	0.956	0.955	0.961	0.962	0.957	0.959	0.951	0.950	0.962	0.945
1467	climate-crashes	540	20	2	0.690	0.731	0.708	0.714	0.417	0.528	0.687	0.748	0.737	0.650
1487	ozone-8hr	2,526	72	2	0.870	0.881	0.901	0.892	0.895	0.899	0.875	0.868	0.897	0.890
4134	bioresponse	3,751	1776	2	0.775	0.774	0.724	0.723	0.768	0.780	–	–	0.785	0.732
46916	coil2000	8,380	85	2	0.654	0.679	0.653	0.662	0.645	0.650	0.634	0.683	0.690	0.660
46924	e-commerce-shipping	10,000	10	2	0.748	0.746	0.746	0.746	0.742	0.744	0.746	0.743	0.746	0.726
46935	hr-analytics	10,000	12	2	0.792	0.794	0.788	0.793	0.783	0.782	0.775	0.774	0.793	0.763
46937	coupon	10,000	24	2	0.708	0.711	0.692	0.697	0.666	0.682	0.659	0.676	0.711	0.707
46947	online-shoppers	10,000	17	2	0.906	0.907	0.909	0.919	0.907	0.907	0.912	0.912	0.913	0.872
46950	polish-bankruptcy	5,850	64	2	0.973	0.975	0.978	0.978	0.948	0.951	0.870	0.874	0.975	0.793
46956	seismic-bumps	2,578	15	2	0.554	0.491	0.755	0.731	0.746	0.749	0.731	0.751	0.740	0.719
PLL, $q = 0.6$ (accuracy)														
42	soybean	631	35	19	0.589	0.798	0.074	0.796	0.388	0.789	–	–	0.833	0.788
307	vowel	990	12	11	0.476	0.586	0.091	0.396	0.217	0.340	–	–	0.607	0.210
1468	cnae-9	1,052	856	9	0.920	0.930	0.141	0.377	0.885	0.892	–	–	0.936	0.841
1478	har	10,000	561	6	0.985	0.984	0.975	0.975	0.969	0.969	–	–	0.982	0.971
1515	micro-mass	571	1300	20	0.548	0.671	0.047	0.221	0.279	0.320	–	–	0.645	0.477
4538	gesture-phase	9,873	32	5	0.507	0.507	0.470	0.470	0.483	0.484	0.488	0.494	0.499	0.496
40499	texture	5,473	40	11	1.000	1.000	0.205	0.999	0.998	0.999	–	–	0.999	0.995
46933	hiva-agnostic	3,825	1617	3	0.964	0.964	0.964	0.964	0.964	0.964	–	–	0.964	0.958

TABLE S12
Study 3 by TFM. G: mean of multi-start − single chain; regr.: pairs counted by C4′′; AUC(H3′′): the diagnostic within the TFM. Last three columns (exploratory): mean difference over datasets and W/T/L.

TFM	datasets	G	regr.	AUC(H3)	multi-TFM − multi-start	multi-start − L	single − L
TabICL v2	20	+0.024	1	0.85	+0.016 (10/5/5)	+0.047 (18/0/2)	+0.024 (16/0/4)
TabPFN-3	20	+0.112	4	0.93	+0.070 (12/5/3)	−0.006 (15/0/5)	−0.118 (11/0/9)
TabDPT	20	+0.037	3	0.59	+0.053 (16/2/2)	+0.011 (14/0/6)	−0.026 (13/0/7)
TabPFN v2	11	+0.013	2	0.64	+0.017 (8/1/2)	+0.023 (8/0/3)	+0.010 (7/0/4)

TABLE S13
The 10 (TFM, dataset) pairs in which the multi-start average is more than 0.02 below the better of the single chain and L (criterion C4′′), ordered by the size of the regression.

TFM	Dataset	single	multi-start	L
TabPFN-3	cnae-9	0.141	0.377	0.841
TabPFN-3	micro-mass	0.047	0.221	0.477
TabICL v2	seismic-bumps	0.554	0.491	0.719
TabDPT	micro-mass	0.279	0.320	0.477
TabDPT	climate-crashes	0.417	0.528	0.650
TabPFN v2	coupon	0.659	0.676	0.707
TabPFN-3	gesture-phase	0.470	0.470	0.496
TabDPT	coupon	0.666	0.682	0.707
TabPFN-3	seismic-bumps	0.755	0.731	0.719
TabPFN v2	ozone-8hr	0.875	0.868	0.890

TABLE S14
Exploratory decomposition of Study 3 (not registered; computed after the confirmatory scoring from the stored predictions, no chain was rerun). Mean test score over datasets (within-TFM rows: mean over the TFMs of each dataset), and difference to the registered average with W/T/L over the 20 datasets and a two-sided Wilcoxon signed-rank p, unadjusted (SciPy defaults: zero differences dropped; exact distribution without zeros, normal approximation otherwise).

Prediction (one TFM)	LLP	PLL	− multi-start	W/T/L	p
single chain (warm start)	0.801	0.589	−0.058	1/6/13	0.002
chain started from L	0.799	0.740	+0.001	5/6/9	0.398
chain with cold start	0.793	0.589	−0.063	0/4/16	<0.001
multi-start (registered: warm, L, cold)	0.807	0.725			
warm and L starts	0.805	0.731	+0.001	5/13/2	0.334
warm and cold starts (no L)	0.806	0.601	−0.050	1/6/13	<0.001
single chain averaged with L's prediction	0.813	0.737	+0.008	12/1/7	0.261
multi-start averaged with L's prediction	0.816	0.765	+0.021	14/2/4	0.005
tuned learner L	0.783	0.717	−0.018	6/0/14	0.143
Prediction (all TFMs)	**LLP**	**PLL**	**− multi-TFM**	**W/T/L**	**p**
multi-TFM (registered)	0.828	0.808			
single chains only, averaged over TFMs	0.823	0.747	−0.028	3/8/9	0.020
warm and cold chains only (no L)	0.829	0.758	−0.020	2/10/8	0.044
chains started from L only	0.823	0.814	−0.001	5/7/8	0.658
without TabPFN v2	0.827	0.808	−0.001	3/13/4	0.286

TABLE S15
Study 3, the two registered differences split by setting (descriptive; one-sided Wilcoxon signed-rank p, unadjusted). The registered tests pool both settings (Table 2).

Hypothesis	Setting (n)	mean	median	W/T/L	p
H1″: multi-start − single chain	LLP (12)	+0.007	+0.002	7/4/1	0.021
H1″: multi-start − single chain	PLL (8)	+0.136	+0.098	6/2/0	0.016
H2″: multi-TFM − random TFM	LLP (12)	+0.021	+0.016	12/0/0	<0.001
H2″: multi-TFM − random TFM	PLL (8)	+0.083	+0.025	6/2/0	0.004

TABLE S16
The three configurations, the evidence for each and their cost. "Confirmatory": a pre-registered test passed; "exploratory": computed after the registered scoring, without error control.

Configuration	Against specialised learners	Against simpler variants	Cost
Single chain, one TFM	confirmatory with TabICL v2 (Study 2); criteria missed with TabPFN v2 (Study 1)	confirmatory: lower log loss than hard-EM	1 chain
Multi-start, one TFM	registered descriptive test not significant (+0.018, $p = 0.14$); 9 of 71 pairs more than 0.02 behind L	confirmatory: above the single chain (H1″); exploratory: no better than the chain started from L	3 chains and tuning L
Multi-TFM	exploratory: above L on 20/0/0 datasets (W/T/L)	confirmatory: above a TFM drawn at random (H2″), level with the best (C5″); exploratory: not significantly above TabICL v2 alone	9 or 12 chains and tuning L

APPENDIX F
SUPPLEMENTARY ANALYSIS S3B: PROTOCOL AND FURTHER RESULTS

Protocol and timeline.: The plan and its scoring script were committed after Study 3 had been scored and before any of its runs, and so with the Study 3 results known; the code that ran was committed afterwards, before any run, and the hashes of all 87 code files in every run match it (Table S2). Three parts: B1, the two non-neural baselines on CPU, with three comparisons registered as descriptive (the multi-TFM average against each baseline, and L against the better of them), two-sided Wilcoxon signed-rank tests Holm-adjusted together; B2, the naive TFM, hard-EM and the oracle with every TFM of Study 3, with one test (single chain against hard-EM on log loss, two-sided); B3, the warm-start chain for 100 sweeps (burn-in 33) on the PLL cells of TabICL v2, TabPFN-3 and TabDPT, with prediction P22. Every row was checked against the split and weak-label hashes of Study 3; none differed.

Deviations.: (i) The plan asked for one scoring on complete data. After the first of the two planned GPU sessions, 5 of the 72 cells of B3 were missing. B1 and B2 were scored; B3 was scored on the one TFM with all its cells, as the plan prescribes for incomplete runs; the 43 available cells of the other two TFMs were then looked at once (recorded in the experiment log); the second session, which had started before that look, added exactly the missing cells, leaving every earlier row unchanged; and B3 was scored again on all 72 cells. At the interim look the mean gain over the available cells was +0.056, so the second part of P22 could have held only if the missing cells had lost more than 0.32 accuracy each on average. (ii) The plan compared L with the better non-neural baseline of each setting; the script took the better of the two on each dataset, which favours the baselines with hindsight. With the baseline of each setting (GBDT for LLP, PL-kNN for PLL) the difference is +0.031 (median -0.002, $9/1/10$, $p_{\mathrm{Holm}}=0.522$); the conclusion is the same. (iii) The plan described GBDT as a default configuration; the code disabled early stopping and relabelled out of fold with 2 folds, as described below.

Baselines.: GBDT self-training uses histogram gradient boosting in one fixed configuration (100 iterations, learning rate 0.1, no early stopping): for PLL, pseudo-labels restricted to each point's candidate set, starting from a random candidate and reassigned out-of-fold (2 folds) for 3 rounds; for LLP, an out-of-fold regression of the bag proportion as the initial score, then 3 rounds in which the top-c_g points of each bag (by current score) are labelled positive. PL-kNN [8] predicts by a vote over the candidate sets of the $k = 10$ nearest training points. Neither was tuned. Across the three seeds, the median over datasets of the standard deviation of every method's score was between 0.013 and 0.027 (Study 3 and B1).

Further results.: The multi-TFM average beat the tree-based learner by +0.135 [+0.051, +0.234] over both settings (median +0.031, $19/0/1$, $p_{\mathrm{Holm}}<0.001$); on PLL the difference (+0.289) mostly reflects the weakness of the tree-based learner, which on 3 of the 8 datasets was within 0.02 of the rate of the majority class or below it (on 1 of them so was the multi-TFM average; Table S19). The single chain was not clearly better than the tree-based learner (+0.031 [-0.003, +0.071], median +0.003, $11/1/8$; not tested): level on LLP (0.801 against 0.796), and on PLL below PL-kNN on average (0.589 against 0.620) because of three datasets on which it stalled (median difference +0.029, $5/0/3$). The tuned learner L was not better than the better untuned baseline of each dataset (+0.022, median -0.003, $7/1/12$, $p_{\mathrm{Holm}}=0.841$); on LLP it was below tree-based self-training on 9 of 12 datasets (-0.013), on PLL above PL-kNN on 7 of 8 (+0.097). In accuracy the single chain was level with hard-EM on LLP (+0.001) and ahead on PLL (+0.050, $5/1/2$), with TabDPT on LLP an exception (hard-EM 0.814 against 0.786). The multi-TFM average stayed 0.085 below the best oracle of each dataset on average (0.052 on LLP, 0.135 on PLL; more than 0.02 below on 12 of 20). With TabPFN-3 on PLL the single chain (0.371) and hard-EM (0.367) stayed near the naive TFM (0.357); the multi-start average reached 0.650, the chain from L 0.692 and L itself 0.717. Prediction P22 (fewer than a quarter of the 35 stalled cells come within 0.02 of the better of the multi-start average and L, and a mean gain below 0.03) was false: 5 cells were repaired (14%), but the mean gain was +0.053. Label accuracy rose by more than 0.03 after sweep 30 in 19 of 72 cells. With TabDPT, runs are not bit-reproducible (12 of its 24 chains differ from the Study 3 chain within the first 30 sweeps).

TABLE S17
B2: mean test score over datasets by TFM (AUC for LLP, accuracy for PLL; number of datasets in brackets).

TFM	Setting (n)	naive	hard-EM	single chain	multi-start	oracle
TabICL v2	LLP (12)	0.707	0.794	0.802	0.803	0.876
TabICL v2	PLL (8)	0.442	0.678	0.749	0.805	0.941
TabPFN-3	LLP (12)	0.729	0.796	0.817	0.817	0.880
TabPFN-3	PLL (8)	0.357	0.367	0.371	0.650	0.940
TabDPT	LLP (12)	0.716	0.814	0.786	0.799	0.874
TabDPT	PLL (8)	0.382	0.577	0.648	0.720	0.925
TabPFN v2	LLP (10)	0.726	0.781	0.784	0.798	0.858
TabPFN v2	PLL (1)	0.447	0.526	0.488	0.494	0.745

APPENDIX G
DEVELOPMENT BETWEEN STUDY 2 AND STUDY 3

The multi-start and multi-TFM samplers were developed between 4 and 10 November 2026 on the datasets of Studies 1 and 2 and on development datasets. All of it is exploratory. We wrote the plan of every run, the analysis script and numbered

TABLE S18
B3: chains run for 100 instead of 30 sweeps (burn-in 33 instead of 10) on PLL ($q = 0.6$). Stuck: the 30-sweep chain was more than 0.02 below the better of the multi-start average and L (Study 3). Repaired: the 100-sweep chain is within 0.02 of that better score. Gain: mean test accuracy after 100 minus after 30 sweeps. Label accuracy: mean over cells of its average over sweeps 28–32 and 96–100, as in the scoring script (at sweeps 30 and 100 exactly: 0.76→0.79, 0.40→0.49 and 0.63→0.65). Last column: cells whose label accuracy rose by more than 0.03 after sweep 30. TabDPT is not bit-reproducible: 12 of its chains differ from the Study 3 chain within the first 30 sweeps (by up to 0.14 in label accuracy).

TFM	cells	stuck	repaired	gain (all)	label acc. ≈30 → ≈100	rose > 0.03 after 30
TabICL v2	24	9	3	+0.031	0.76 → 0.78	5
TabPFN-3	24	16	1	+0.111	0.40 → 0.49	10
TabDPT	24	10	1	+0.016	0.63 → 0.65	4
All	72	35	5	+0.053		19

TABLE S19
B1 per dataset (mean over 3 seeds; TFM-based columns averaged over the TFMs of each dataset). "Majority": accuracy of always predicting the most frequent class of the test split (PLL).

Setting	Dataset	majority	GBDT	PL-kNN	single	multi-start	L	multi-TFM
LLP	1038	–	0.949	–	0.979	0.980	0.939	0.988
LLP	1046	–	0.959	–	0.956	0.957	0.945	0.962
LLP	1467	–	0.654	–	0.625	0.680	0.650	0.737
LLP	1487	–	0.893	–	0.885	0.885	0.890	0.897
LLP	4134	–	0.763	–	0.756	0.759	0.732	0.785
LLP	46916	–	0.638	–	0.646	0.669	0.660	0.690
LLP	46924	–	0.743	–	0.745	0.745	0.726	0.746
LLP	46935	–	0.781	–	0.785	0.786	0.763	0.793
LLP	46937	–	0.708	–	0.681	0.691	0.707	0.711
LLP	46947	–	0.908	–	0.909	0.911	0.872	0.913
LLP	46950	–	0.934	–	0.942	0.944	0.793	0.975
LLP	46956	–	0.621	–	0.697	0.681	0.719	0.740
PLL	42	0.142	0.130	0.679	0.350	0.795	0.788	0.833
PLL	307	0.091	0.270	0.365	0.261	0.441	0.210	0.607
PLL	1468	0.114	0.373	0.522	0.649	0.733	0.841	0.936
PLL	1478	0.189	0.966	0.878	0.976	0.976	0.971	0.982
PLL	1515	0.105	0.122	0.209	0.291	0.404	0.477	0.645
PLL	4538	0.299	0.534	0.448	0.487	0.489	0.496	0.499
PLL	40499	0.091	0.799	0.915	0.734	0.999	0.995	0.999
PLL	46933	0.964	0.959	0.945	0.964	0.964	0.958	0.964

predictions with thresholds before the run; Table S20 lists every prediction and its outcome (P10 and P19–P21 were not used; P22 is in Appendix F).

TABLE S20
Predictions written before each development run, and outcomes. "Regression": more than 0.02 below the better of the single chain and the tuned learner L. P1–P4 belong to a first phase in which one chain was selected by weak-label fit; P4 failed and the selection rule was replaced by averaging.

	Prediction (written before the run)	Result	Outcome
P1	a chain of TabPFN v2 started from tuned CC ends above 0.80 label accuracy in 9/9 runs and is within 0.02 of CC	0.95–0.99; above CC on 3/3 datasets	correct
P2	on semeion, where the gain is negative at 0.85–0.95, that chain still ends below 0.50	ends at 0.76–0.81	wrong
P3	LLP, TabPFN v2: where the gain is negative at the start, the chain ends below its start (2 datasets)	2/2 datasets	correct
P4	choosing the chain by weak-label fit picks a near-best start in $\geq 80\%$ of cells	19/24 (79.2%)	wrong
P5	stacking with L has fewer regressions than equal averaging (TabPFN v2)	1 regression each	wrong
P6	aggregators differ by < 0.005 in mean gain (TabICL v2)	max. gap 0.0031	correct
P7	the chosen average is not below the single chain by more than 0.002 in any of four groups	+0.0019 to +0.0043	correct
P8	chain-seeded map predicts final label accuracy: median error ≤ 0.10	0.016 (234 cells)	correct
P9	chain-seeded map is more accurate than the uniform map	0.016 vs. 0.073	correct
P11	TabPFN-3 has no low attractor on the four stalling datasets	0/4 datasets	wrong
P12	multi-start at equal cost (3×10 sweeps) still gains, ≤ 3 regressions	6 regressions	wrong
P13	four-TFM average within 0.01 of the best TFM on average, > 0.02 behind on ≤ 2 datasets (seed 0)	-0.008; 2 worse by > 0.02	correct
P14	TabPFN-3 multi-start beats TabPFN v2 multi-start by 0.05 on $\geq 3/4$ stalling datasets	1/4	wrong
P15	disagreement of three 3-sweep probe chains predicts basin splits (AUC ≥ 0.75)	AUC 0.45	wrong
P16	same on 23 datasets, 3 seeds (≤ 3 behind), and not below any single TFM on average	-0.004; 1 worse by > 0.02	correct
P17	multi-start gains with TabPFN-3 and TabDPT, ≤ 2 regressions each	+0.0115 / +0.0002; 2 / 2	correct
P18	post-chain diagnostic: AUC ≥ 0.80 on new development cells	AUC 0.86 (138 cells)	correct

Choice of the average.: Three ways of combining the three chains were compared on 78 (TFM, dataset) pairs (234 cells; TabICL v2 and TabPFN v2; four LLP datasets of TabPFN v2 are missing because of download errors): equal weights, and two stacking rules that use weak-label fit. The registered rule chose the combination with the fewest regressions: equal weights (2 against 3 and 3). On these pairs the equal-weight average gained +0.0032 over the single chain (mean of the four TFM–setting group means; better on 39, worse on 11, tied on 28) and was never more than 0.02 below it. This small gain is why we expected a typical dataset of Study 3 to gain little, which held (median +0.003), and few pairs to regress, which did not. Running the three chains for a third of the sweeps each, at the cost of one chain, lost the gain (P12, scored on the 145 of

234 cells that finished).

TFMs.: TabPFN-3 was preferred to its successor TabPFN-3.5 by a registered rule on speed (the latter was 3.1 times slower per call on a T4, above a registered limit of 1.5). Under uniformly placed errors TabPFN-3 keeps the low attractor of TabPFN v2 on the four stalling datasets of Study 1 (P11), yet its chains started from L reach high accuracy on them, and so do those of TabPFN v2 on three of the four (P14 was wrong for that reason): stalling is a matter of the start as much as of the TFM. On 23 development and earlier datasets the four-TFM average was -0.004 from the best TFM of each dataset (P16), which fixed criterion C5″; P13, on an earlier run, was scored on the 13 of 15 datasets with all four TFMs.

Response maps.: For every cell of the earlier runs we measured the response to redrawing every block once, at a grid of accuracy levels, placing errors either uniformly or according to the stored label marginals of the chain (wrong points chosen with probability proportional to one minus the marginal of the true label), and predicted the final label accuracy by a rule fixed in advance: start at the chain's initial accuracy and follow the sign of the interpolated gain to the next crossing. Table S21 and Figure S4 give the errors. The 90th percentile of the error with chain-seeded maps is 0.051. The uniform-error map is not a faithful model of the chains: with TabPFN v2, chains on semeion, within the scope of Theorem 3, climbed from 0.17 after one sweep to 0.34–0.37, inside the region of negative gain, and chains on optdigits (subsampled context; 2 of 3 seeds) climbed through it. On 234 cells of development runs, chains ended above the fixed point of the uniform-error map in 96% of cells (median error 0.073), and in 94% of the 108 PLL cells, to which Assumption 1 applies: a chain's own errors sit on points the TFM finds ambiguous and harm the context less than uniform ones, so for the uniform map only the lower half of Assumption 1 is plausible. With errors placed according to the chain's own marginals the fixed point is within a median of 0.016 of the final accuracy (Figure S4); those marginals come from the chain whose end is predicted, so this shows that the placement of errors matters, not that the map forecasts anything. Comparing the two maps at equal r bounds the tolerance of Assumption 1 from below: in the 78 PLL cells within the theorem's scope, $\delta \geq 0.05$ in the median cell and up to 0.14 (Section A.6). Of three predictions about starts written in advance, two held (started from tuned CC, TabPFN v2 chains ended at label accuracies 0.95–0.99 on three of the datasets); one was wrong. On the 3 datasets within the theorem's scope (n_F from 756 to 1,395) a floor over 30 sweeps needs a gain margin of 0.25 to 0.35 even with $\delta = 0$, and the measured gains are at most 0.14.

TABLE S21
Median absolute error of the predicted final label accuracy, by the way errors are placed when the response map is measured ("structured": according to the warm start, PLL only).

TFM	Setting	cells	chain-seeded	uniform	structured
TabICL v2	LLP	96	0.026	0.081	–
TabICL v2	PLL	69	0.011	0.042	0.053
TabPFN v2	LLP	30	0.025	0.070	–
TabPFN v2	PLL	39	0.009	0.083	0.079
All		234	0.016	0.073	

Diagnostics.: A first label-free signal, the disagreement of three short probe chains, carried no information (P15). The second, the disagreement of the warm-start and learner-start chains, was proposed on old runs, held on new development cells (P18, AUC 0.86 on 138 cells) and then failed its registered threshold in Study 3 (H3″).

Harder settings, not pursued.: With bags of 32 every TFM had cells in which the chain converged to exchanged classes, and the multi-start average repaired them only in part. Candidate sets whose distractors depend on the instance (drawn in proportion to a classifier's probabilities), run with TabICL v2 only, were much harder than uniform ones. Bags formed by clustering the features were easier than random bags and were dropped. These settings are left to future work.

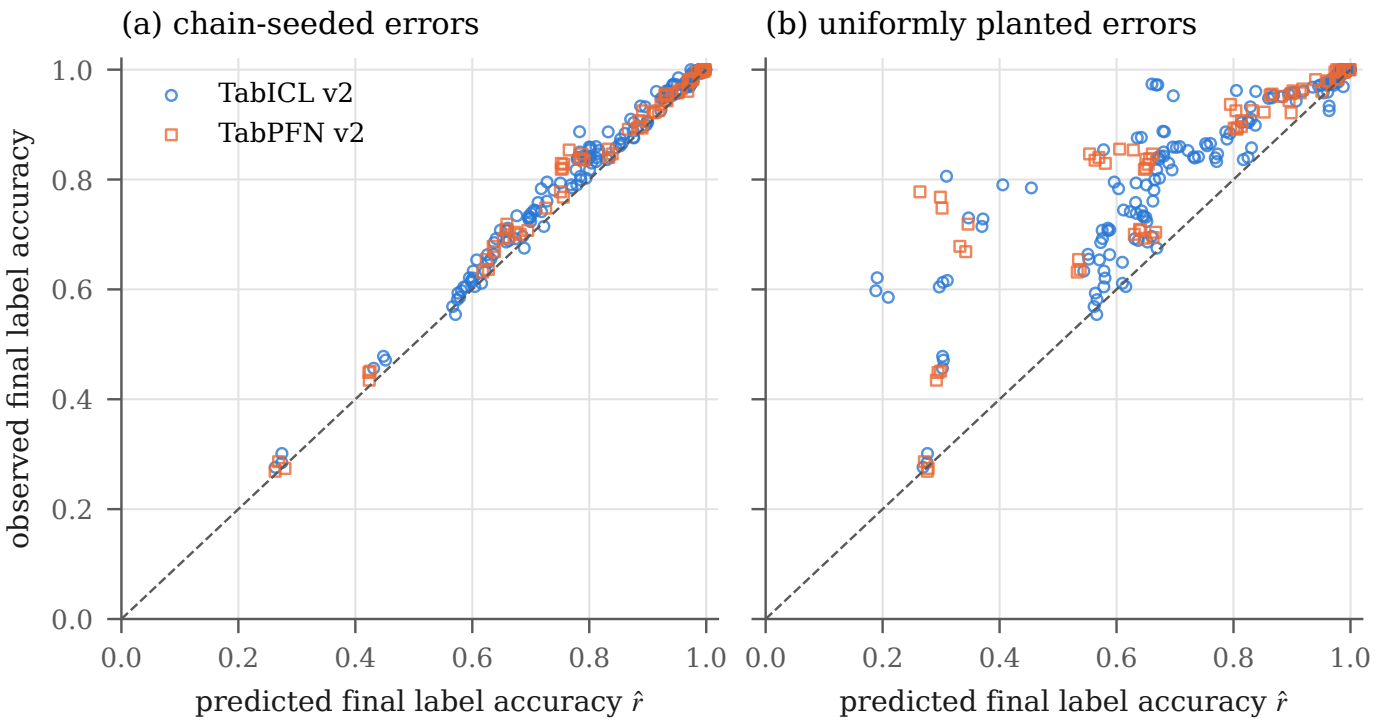


Fig. S4. Final label accuracy of the warm-start chains of 234 development cells against the fixed point predicted from a response map, with errors placed (a) by the chain's own label marginals, (b) uniformly.

APPENDIX H
POSITIONING AGAINST THE CLOSEST WORKS

TABLE S22
Positioning against the closest works. "Partial": exact posterior over labels inside an EM loop, with conditionally independent predictions. *The TFM is frozen; the LLP warm start trains a linear DLLP model and the learner start a tuned specialised learner (Section 4). †The default uncorrected-block sampler has only the prediction-level analysis of Proposition S3.

	no training (frozen model)	tabular data	several weak-label types	inference over latent labels with guarantees	dependence among latent labels
In-Context PU [9]	– (pretrains a PFN)	✓	– (PU only)	–	–
GLWS [10], ILL [11]	–	–	✓	partial	–
TabPFN imputation [12]	✓	✓	– (missing outcome)	–	–
Noise immunity [13]	✓	✓	– (label noise)	– (measurement)	–
In-Context Gibbs (ours)	✓*	✓	✓	✓ (exact variants only)†	✓

APPENDIX I
CORRECTNESS EXPERIMENTS

We enumerate every labelling of small problems ($n \leq 14$), build the exact transition matrix of each kernel with an analytic PPD as $\tilde{p}$, and compare its stationary distribution with π. Two priors are used: one in which features carry no information about the label mechanism, and a three-component mixture in which they do. Across all 14 configurations the TV distance is at most 3×10^{-14} (Table S23), and running the actual sampler for 10^5 sweeps reaches the Monte Carlo floor of i.i.d. sampling. For Theorem 2 we use conditionals with deliberate, incompatible perturbations (30 configurations) and small PFNs trained on the same prior; the bound held in 42 of 42 configurations and was 1.9 to 15 times loose (median 3.9; Table S24); a fully trained PFN gives TV 0.012 (single-site) and 0.007 (swap). On an exactly solvable pooled problem with $n = 1{,}000$ and $K = 10$, blocking over all points changes predictions by 1.7×10^{-4}, whereas blocking only the free points biases them by 0.031, which is why blocks include fixed points.

TABLE S23
Exactness (Theorem 1, Proposition S2(a)): TV between the stationary distribution of the exact kernel and π.

Prior	Weak labels	Kernel	n	states	TV(stationary, π)
cell	PU-SCAR (c=0.5)	single	10	128	3.3e-15
cell	SSL	single	10	128	4.0e-15
cell	PLL	single	10	128	4.0e-15
cell	Noisy (T known)	single	8	256	1.9e-14
cell	LLP (3 bags of 4)	swap	12	96	3.1e-15
cell	SSL, block-MH (m=3)	block-mh	10	128	2.8e-14
mixture	PU-SCAR (c=0.5)	single	10	128	3.3e-15
mixture	SSL	single	10	128	3.0e-15
mixture	PLL	single	10	128	3.0e-15
mixture	Noisy (T known)	single	8	256	2.3e-14
mixture	LLP (3 bags of 4)	swap	12	96	4.8e-15
mixture	SSL, block-MH (m=3)	block-mh	10	128	4.0e-15
cell	PLL (C=3, |S_i|=2)	single	7	128	1.2e-14
mixture	PLL (C=3, |S_i|=2)	single	7	128	3.8e-15

TABLE S24
Stability (Theorem 2): actual TV between the stationary distributions of the approximate and exact chains, the bound $\tau(P)\bar{\varepsilon}$, and their ratio (looseness).

Conditionals	Kernel	configs	median TV($\tilde{\pi}, \pi$)	median bound	bound held	median looseness
perturbed PPD (incompatible)	single	15	0.0181	0.246	15/15	9.8
perturbed PPD (incompatible)	swap	15	0.0254	0.066	15/15	2.2
small PFN (trained)	single	6	0.1419	0.628	6/6	4.9
small PFN (trained)	swap	6	0.0480	0.107	6/6	3.3

APPENDIX J
ABLATIONS

Exact sampler.: The MH-corrected block sampler, exact when the TFM is and the context is not subsampled, was not consistently better than the default at the same 30 sweeps (MH minus default with TabICL v2: +0.033 AUC on LLP, -0.021 accuracy on PLL at $q = 0.6$; Table S25) and was a median 7 times slower; the comparison confounds bias with slower mixing. Chains do not mix at the level of individual labels (Table S26).

TABLE S25
Mixing and the exact sampler, by TFM. TabPFN: Study 1 (registered ablations); TabICL: exploratory lot on the Study 1 datasets. "3 small": the three smallest datasets of each group, as registered for $\widehat{R}$. Columns 2–5: medians over cells of the fraction of training labels with $\widehat{R} > 1.1$ (two chains) and of the absolute difference between the test metrics of the two chains. Columns 6–7: mean difference in AUC (LLP) or accuracy (PLL) between the MH-corrected and the default sampler at 30 sweeps, with the mean MH acceptance rate. In the last row, $\widehat{R}$ uses all four datasets with $\geq$8 classes, and the MH comparison only two of them (mfeat-factors, optdigits; 5 of 12 cells), because the corrected sampler ran out of the 12-hour compute session on optdigits.

	labels with $\widehat{R} > 1.1$		test-metric gap, 2 chains		MH-corrected − default	
Setting (datasets)	TabPFN	TabICL	TabPFN	TabICL	TabPFN	TabICL [acc. rate]
LLP, bag 8 (3 small)	11%	28%	0.041	0.018	+0.010	+0.033 [0.68]
PLL, $q = 0.3$ (3 small)	6%	6%	0.006	0.018	+0.006	−0.009 [0.78]
PLL, $q = 0.6$ (3 small)	21%	31%	0.016	0.021	–	−0.021 [0.72]
PLL, $q = 0.6$, $\geq$8 classes (4; MH: 2)	–	22%	–	0.002	–	−0.016 [0.47]

TABLE S26
Study 2 confirmatory tests with a second chain per cell (exploratory lot run after the analysis; same data, weak labels and start, other sampling seed). Each entry: mean difference, W/T/L over datasets, p_{Holm} (Holm within each hypothesis, as registered). Chain a is the registered result. The averaged column uses the stored test predictions of both chains (float16).

Test	chain a (registered)	chain b	chains a and b averaged
H2′: LLP, bag 8, vs DLLP (linear)	+0.052, 16/0/2, <0.001	+0.046, 15/0/3, 0.002	+0.052, 17/0/1, <0.001
H4′: PLL, $q = 0.3$, vs CC (tuned)	+0.012, 8/2/0, 0.016	+0.013, 9/1/0, 0.008	+0.013, 9/1/0, 0.008
H4′: PLL, $q = 0.6$, vs CC (tuned)	+0.012, 8/1/1, 0.016	+0.011, 8/1/1, 0.074	+0.013, 8/1/1, 0.008

TABLE S27
In-Context Gibbs versus hard-EM with the same TFM and warm start, log loss. A cell is a (dataset, weak-label level) pair; Bayesian signed-rank test with ROPE ± 0.005; Spearman correlation between weakness level and the gap. †Computed on PLL cells (LLP has a single level), as stated before the analysis was run.

Study (TFM; settings)	cells	Gibbs better	P(Gibbs better)	P(ROPE)	Spearman ρ
Study 1 (TabPFN v2; all six settings)	128	86.7%	1.00	0.00	0.36
Study 2 (TabICL v2; LLP bag 8, PLL)	38	78.9%	1.00	0.00	0.66†

TABLE S28
Study 1 ablations (TabPFN v2): each variant against the default on the same cells. Blocks, sweeps and MH: the three smallest datasets per group; transductive and cold start: all datasets.

Variant (vs. default)	Setting	cells	Δ AUC/acc.	W/T/L	Δ log loss (+ = better)
$K = 20$	LLP, bag 8	9	+0.007	5/0/4	+0.024
$K = 20$	PLL, 0.3	9	+0.006	5/2/2	−0.000
$K = 5$ (default 10)	LLP, bag 8	9	−0.005	6/0/3	+0.006
$K = 5$ (default 10)	PLL, 0.3	9	+0.004	5/1/3	+0.009
15 sweeps (default 30)	LLP, bag 8	9	+0.005	4/0/5	+0.001
15 sweeps (default 30)	PLL, 0.3	9	+0.004	4/2/3	−0.001
60 sweeps	LLP, bag 8	9	+0.014	4/0/5	−0.012
60 sweeps	PLL, 0.3	9	+0.005	4/5/0	+0.009
MH-corrected blocks (exact)	LLP, bag 8	9	+0.010	4/0/5	−0.019
MH-corrected blocks (exact)	PLL, 0.3	9	+0.006	4/3/2	+0.009
transductive prediction	LLP, bag 8	30	−0.013	7/0/23	−0.042
cold start	LLP, bag 8	30	−0.005	10/0/20	−0.002
cold start	PLL, 0.3	42	+0.000	11/23/8	+0.000
cold start	PLL, 0.6	42	+0.012	14/9/19	+0.023
PU-SAR prior $N(0, 1)$	PU-SAR, 0.1	30	−0.002	15/3/12	−0.015
PU-SAR prior $N(0, 1)$	PU-SAR, 0.3	30	+0.001	10/1/19	+0.002
PU-SAR prior $N(0, 4^2)$	PU-SAR, 0.1	30	−0.004	7/9/14	+0.000
PU-SAR prior $N(0, 4^2)$	PU-SAR, 0.3	30	−0.001	10/12/8	−0.016

APPENDIX K
REAL PARTIAL LABELS

TABLE S29
Real partial labels (Study 1, TabICL v2 for all TFM-based methods): accuracy averaged over 10 random 90/10 splits, and mean rank over all splits of the three datasets among the ten ranked methods (In-Context Gibbs, naive TFM and eight baselines). Bold: best non-oracle accuracy.

Method	BirdSong	Lost	MSRCv2	mean rank
In-Context Gibbs (ours)	**0.759**	**0.750**	**0.628**	2.15
Hard-EM (same TFM)	0.730	0.665	0.627	–
Stochastic EM (same TFM)	0.744	0.716	0.622	–
Naive TFM	0.689	0.481	0.491	7.27
PRODEN (default)	0.667	0.373	0.344	9.88
PRODEN (tuned)	0.707	0.748	0.465	5.12
GLWS (default)	0.722	0.712	0.582	3.98
GLWS (tuned)	0.731	0.747	0.490	4.03
RC (default)	0.719	0.720	0.424	6.00
RC (tuned)	0.702	0.721	0.478	5.23
CC (default)	0.719	0.717	0.425	6.02
CC (tuned)	0.700	0.723	0.479	5.32
Oracle (true labels)	0.835	0.889	0.824	–

APPENDIX L
DATASETS, BASELINES AND TUNING

Datasets.: Study 1 uses every OpenML-CC18 dataset with real (non-synthetic) data, 500 to 10,000 rows, at most 500 features and, for binary tasks, a minority class of at least 10% (multiclass: 3–10 classes), one per source; Study 2 applies the same criteria to five curated suites after excluding OpenML-CC18 and Study 1 (Table S31); the selection for Study 3 is described in Appendix E. Duplicate rows are removed before splitting. TFMs receive ordinally encoded features; MLP baselines receive median-imputed, standardised features with one-hot encoded categories, fitted on the training split.

Baselines and tuning.: Each baseline runs in the configuration of its original code or paper and tuned by 20 random configurations (the default among them) from the spaces below, scored by the method's own loss on a 20% validation split with weak labels only (split by bag for LLP); the best configuration is refit on the full training split. MLP-based learners (DLLP, LLP-VAT, EasyLLP, GLWS, PRODEN) search learning rate $\sim \mathrm{LogU}(10^{-4}, 10^{-2})$, weight decay $\sim \mathrm{LogU}(10^{-6}, 10^{-2})$, hidden layers in $\{(), (64), (128, 128)\}$ and epochs in $\{100, 200, 400\}$, plus bags per batch for LLP, VAT parameters for LLP-VAT, the optimiser for PRODEN and the entropy term for GLWS/ILL; RC and CC use a linear model with the PRODEN space (Table S30).

TABLE S30
Search spaces (20 random configurations, the default included). LogU: log-uniform.

Baseline	Default	Search space
DLLP (MLP, linear), EasyLLP	lr 10^{-3}, wd 0, 200 epochs	lr $\sim$ LogU$(10^{-4}, 10^{-2})$; wd $\sim$ LogU$(10^{-6}, 10^{-2})$; hidden $\in \{(), (64), (128, 128)\}$; epochs $\in \{100, 200, 400\}$; bags/batch $\in \{8, 32\}$
LLP-VAT	as DLLP + VAT defaults of the official code	as DLLP; $\xi \in \{10^{-6}, 10^{-3}\}$; $\epsilon \in \{0.1, 0.5, 1, 2\}$; consistency weight $\sim$ LogU$(10^{-2}, 1)$
GLWS / ILL	official code, MLP backbone	lr, wd as DLLP; hidden $\in \{(64), (128), (128, 128)\}$; epochs $\in \{100, 200, 400\}$; entropy term on/off (PLL)
PRODEN	lr 10^{-3}, wd 10^{-5}, hidden $(128, 128)$, 200 epochs, SGD	lr, wd, hidden, epochs as DLLP; optimiser $\in$ {SGD (momentum 0.9), Adam}
RC, CC	linear model as in the original paper	as PRODEN
SAR-EM (PU-SAR)	$C_f = C_e = 1$	$C_f, C_e \sim$ LogU$(10^{-3}, 10^{3})$
nnPU, uPU, forward correction	lr 10^{-3}, wd 0, hidden (100), 200 epochs	lr $\sim$ LogU$(10^{-4}, 10^{-2})$; wd $\sim$ LogU$(10^{-6}, 10^{-2})$; hidden $\in \{(64), (128), (128, 128)\}$; epochs $\in \{100, 200, 400\}$
VPU	official code	lr $\sim$ LogU$(10^{-5}, 10^{-3})$; mixup $\lambda \in \{0.01, 0.03, 0.1, 0.3\}$; $\alpha \in \{0.1, 0.3, 1\}$
Self-training	logistic regression, $C = 1$, threshold 0.75	$C \sim$ LogU$(10^{-3}, 10^{3})$; threshold $\in \{0.6, 0.75, 0.9, 0.95\}$
Label spreading	kNN, $k = 7$, $\alpha = 0.2$	kernel $\in$ {kNN, RBF}; $k \in \{3, 5, 7, 11, 15, 21\}$; $\gamma \sim$ LogU$(10^{-3}, 10)$; $\alpha \in \{0.1, 0.2, 0.5, 0.8\}$
VIME	official code	$p_m \in \{0.1, 0.3, 0.5\}$; $\alpha \in \{0.5, 1, 2\}$; $\beta \in \{0.1, 1, 10\}$; hidden $\in \{(64), (128)\}$
Confident learning + TFM	default (not tuned)	–

TABLE S31
Datasets (n after removing duplicates; *subsampled to 10,000 rows by the registered rule; S1: flagged for the sensitivity analysis).

Study 1 (OpenML-CC18)

ID	name	n	d	C	S1	ID	name	n	d	C	S1
3	kr-vs-kp	3,196	36	2		28	optdigits	5,620	64	10	
29	credit-approval	690	15	2		46	splice	3,006	60	3	
44	spambase	4,210	57	2		182	satimage	6,430	36	6	
50	tic-tac-toe	958	9	2		188	eucalyptus	736	19	5	
1049	pc4	1,344	37	2		458	analcatdata_authorship	841	70	4	
1489	phoneme	5,395	5	2		469	analcatdata_dmft	556	4	6	
1494	qsar-biodeg	1,052	41	2		1475	first-order-theorem-proving	5,636	51	6	
6332	cylinder-bands	540	37	2		1497	wall-robot-navigation	5,456	24	4	✓
23381	dresses-sales	500	12	2		1501	semeion	1,593	256	10	
38	sick	3,711	29	2		40966	MiceProtein	1,080	77	8	✓
11	balance-scale	625	4	3	✓	40975	car	1,728	6	4	✓
12	mfeat-factors	1,994	216	10		40982	steel-plates-fault	1,941	27	7	

Study 2 (new curated suites)

ID	name	n	d	C	S1	ID	name	n	d	C	S1
24	mushroom	8,124	22	2		46930	hazelnut-spread-contaminant-detection	2,400	30	2	
312	scene	2,400	299	2		46938	Is-this-a-good-customer	1,723	13	2	
470	profb	672	9	2		46940	Marketing_Campaign	2,058	25	2	
934	socmob	1,156	5	2		46969	NATICUSdroid	7,491	86	2	
40536	SpeedDating	8,378	120	2	✓	181	yeast	1,453	8	10	
41143	jasmine	2,984	144	2		375	JapaneseVowels	9,961	14	9	✓
41145	philippine	5,832	308	2		1466	cardiotocography	2,115	35	10	
41146	sylvine	5,124	20	2		23380	cjs	1,811	33	6	✓
41156	ada	4,146	48	2		40498	wine-quality-white	3,961	11	7	
44130	eye_movements	7,608	20	2	✓	46906	anneal	886	38	5	
45026	heloc	9,444	22	2		46960	students_dropout_and_academic_success	4,424	36	3	
45039	compas-two-years	2,712	11	2		46963	website_phishing	724	9	3	
46911	Bank_Customer_Churn	10,000	10	2		46980	MIC	1,699	111	8	
46927	Fitness_Club	1,499	6	2		1476	gas-drift	10,000*	128	6	✓

APPENDIX M
SECONDARY SETTINGS OF STUDY 1

Study 1 also registered secondary settings (Table S32). In-Context Gibbs improved strongly on the naive TFM for larger bags, as for bags of 8, but for PU with labels selected completely at random, class-dependent noise and SSL it was worse than the naive TFM at 6 of 6 levels (by up to -0.054 AUC, under class-dependent noise at $r = 0.4$). This is the pattern Proposition S5 predicts; the losses may also reflect the bias of uncorrected blocks, which shrinks label counts towards the prior (Proposition S3). The pattern was registered in advance as exploratory analyses D1–D3: the advantage over the naive TFM grows from $q = 0.3$ to $q = 0.6$ (D1, one-sided $p = 0.039$) and from bags of 8 to 16 (D2, $p = 0.010$), and stays below $+0.02$ AUC in the settings where the naive labels lose no information (D3, all six levels); D3 is also met by a method that is worse than the naive TFM, as In-Context Gibbs was. Against the primary comparators of these settings no difference was significant except class-dependent noise at $r = 0.4$, where confident learning combined with the TFM was better.

TABLE S32
Study 1 secondary settings (TabPFN v2). Δ positive favours In-Context Gibbs. †Log loss after matching the known class prior (sign flipped so that positive is better).

Setting	Metric	Primary comparator	In-Context Gibbs − comparator			In-Context Gibbs − naive TFM	
			Δ	W/T/L	p_{Holm}	Δ	W/T/L
LLP, bag 16	AUC	DLLP (linear)	+0.040	7/0/3	0.232	+0.206	9/0/1
LLP, bag 32	AUC	DLLP (linear)	+0.048	6/0/4	0.375	+0.223	10/0/0
PU-SAR, $\mathbb{E}[e \mid y = 1] = 0.1$	AUC	nnPU	+0.034	6/0/4	0.967	+0.007	6/0/4
PU-SAR, $\mathbb{E}[e \mid y = 1] = 0.3$	AUC	nnPU	+0.032	8/0/2	0.109	+0.017	6/0/4
PU-SCAR, $c = 0.1$	log loss†	VPU, tuned	+0.118	8/0/2	0.111	−0.033	5/0/5
PU-SCAR, $c = 0.3$	log loss†	VPU, tuned	+0.179	8/0/2	0.111	−0.028	2/0/8
Class-dependent noise, $r = 0.2$	AUC	CL + TFM	−0.006	3/0/7	0.105	−0.013	1/0/9
Class-dependent noise, $r = 0.4$	AUC	CL + TFM	−0.048	2/0/8	0.020	−0.054	1/0/9
SSL, 5% labelled	AUC	Self-training, tuned	−0.002	5/0/5	1.000	−0.041	0/0/10
SSL, 20% labelled	AUC	Self-training, tuned	−0.009	8/0/2	0.750	−0.016	0/0/10

APPENDIX N
STATEMENTS

Data and code availability.: All datasets are public: OpenML (dataset identifiers in Appendices E and L) and the PALM collection of real partial-label datasets. A further supplementary archive contains the scoring scripts of the three studies and of the supplementary analysis, the list of freeze events with the hashes of the frozen files, and SHA-256 hashes of the result files. Code, frozen protocols, raw per-cell results with code and data hashes, and the scripts that generate every number, table and figure of this paper will be released with the paper.

Licences.: TabICL: BSD-3-Clause; TabDPT and the `tabpfn` package: Apache-2.0 (as declared in their package metadata). The TabPFN v2 and TabPFN-3 weights are distributed by their publisher under its own terms, which the user accepts before download for TabPFN-3; we used all checkpoints for research only and redistribute none of them. OpenML datasets are used under the licence listed on each dataset's page; the PALM datasets as distributed by their authors; we redistribute no dataset. Baseline code without a licence is loaded at run time and not redistributed.

Compute.: All runs used free Kaggle sessions with two NVIDIA T4 GPUs (TFM-based methods) or CPUs (baselines). Study 1 used about 42 hours of two-GPU sessions and 26 hours of four-process CPU sessions; Study 2 about 4 GPU-hours and 80 CPU process-hours; the exploratory lots of Studies 1 and 2 about 13 hours of two-GPU sessions; Study 3 three two-GPU sessions of 8 to 11 hours; the development runs of Appendix G six further sessions of at most 11 hours; the supplementary analysis two two-GPU sessions (11 and 4.5 hours) and one CPU session of about one hour. Development and pilot runs before each freeze are not included.

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REFERENCES

[1] S. Müller, N. Hollmann, S. Pineda Arango, J. Grabocka, and F. Hutter, "Transformers can do Bayesian inference," in *International Conference on Learning Representations*, 2022.

[2] P. J. Schweitzer, "Perturbation theory and finite Markov chains," *Journal of Applied Probability*, vol. 5, no. 2, pp. 401–413, 1968.

[3] C. D. Meyer, Jr., "The condition of a finite Markov chain and perturbation bounds for the limiting probabilities," *SIAM Journal on Algebraic and Discrete Methods*, vol. 1, no. 3, pp. 273–283, 1980.

[4] E. Seneta, "Perturbation of the stationary distribution measured by ergodicity coefficients," *Advances in Applied Probability*, vol. 20, no. 1, pp. 228–230, 1988.

[5] G. E. Cho and C. D. Meyer, "Comparison of perturbation bounds for the stationary distribution of a Markov chain," *Linear Algebra and its Applications*, vol. 335, no. 1–3, pp. 137–150, 2001.

[6] W. Hoeffding, "Probability inequalities for sums of bounded random variables," *Journal of the American Statistical Association*, vol. 58, no. 301, pp. 13–30, 1963.

[7] S. Agarwal, "Surrogate regret bounds for bipartite ranking via strongly proper losses," *Journal of Machine Learning Research*, vol. 15, no. 49, pp. 1653–1674, 2014.

[8] E. Hüllermeier and J. Beringer, "Learning from ambiguously labeled examples," *Intelligent Data Analysis*, vol. 10, no. 5, pp. 419–439, 2006.

[9] S. Liu, Y. Chang, M. Cheng, Q. Tian, and P. Li, "In-context positive-unlabeled learning," *arXiv preprint arXiv:2605.05591*, 2026.

[10] H. Chen, J. Wang, L. Feng, X. Li, Y. Wang, X. Xie, M. Sugiyama, R. Singh, and B. Raj, "A general framework for learning from weak supervision," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235, 2024, pp. 7462–7485.

[11] H. Chen, A. Shah, J. Wang, R. Tao, Y. Wang, X. Li, X. Xie, M. Sugiyama, R. Singh, and B. Raj, "Imprecise label learning: A unified framework for learning with various imprecise label configurations," in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 59 621–59 654.

[12] J. Sepin, "Multiple imputation of a continuous outcome with fully observed predictors using TabPFN," *Stats*, vol. 9, no. 2, p. 38, 2026.

[13] J. Hu and M. Ghelichi, "Noise immunity in in-context tabular learning: An empirical robustness analysis of TabPFN's attention mechanisms," *arXiv preprint arXiv:2604.04868*, 2026.